Stochastic and Sequential Regimes: Extending the Decision Quotient to Dynamic Information Sufficiency

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Abstract

We extend the static decision quotient framework of Paper 4 to stochastic and sequential regimes. Given a decision problem with actions A, factored state space $S = X_1 \times \cdots \times X_n$, and utility U, we ask: what complexity is required to identify sufficient coordinate sets when (a) the state is drawn from a distribution P, or (b) decisions unfold over time with transitions and observations?

Results:

- **Stochastic regime**: STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential regime**: SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy**: Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.
- **Substrate cost**: The integrity-competence verdict is substrate-independent; trajectories diverge based on substrate cost κ.
- **Temporal learning**: Bayesian structure detection reduces abstention over time.
- **Temporal integrity**: Evidence-monotone retractions preserve integrity across sequences.

The reduction constructions are machine-checked in Lean 4 (3288 lines, 124 theorems).

Keywords: computational complexity, decision theory, stochastic decision problems, POMDPs, polynomial hierarchy, PSPACE

1 Introduction

Consider any decision problem $\mathcal{D}$ with actions A and factored state space $S = X_1 \times \cdots \times X_n$. A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sufficient* if knowing only coordinates in I determines the optimal action:

$$s_I = s'_I \implies \mathrm{Opt}(s) = \mathrm{Opt}(s')$$

This is *coordinate sufficiency*—the formalization of *information sufficiency* (whether available information determines the optimal action) over a factored state space.

Paper 4 established that SUFFICIENCY-CHECK is coNP-complete for static decision problems (A, S, U). This paper extends that landscape to two natural extensions: stochastic and sequential regimes.

1.1 Stochastic Regime

When the state is drawn from a distribution $P : \Delta(S)$, sufficiency must account for randomness. We use the distribution-conditional definition: I is stochastically sufficient if $P(s \mid s_I) \otimes U$ determines the same optimal action as $P(s) \otimes U$. This subsumes expected-value sufficiency (where only the first moment matters) and connects to classical sufficient statistics.

Regime	Tuple	Complexity
Static	(A, S, U)	coNP-complete (Paper 4)
Stochastic	(A, S, U, P)	PP-complete
Sequential	(A, S, U, T, O)	PSPACE-complete

The PPupper bound comes from counting satisfying assignments weighted by P. The lower bound uses a MAJSAT reduction: given φ, construct uniform P over $\{0,1\}^n$ and let $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$. Then $\emptyset$ is stochastically sufficient iff $\mathbb{E}_P[\varphi(s)] \geq 1/2$ or $< 1/2$—i.e., majority of assignments satisfy φ.

Tractable subcases: product distributions (expected utility decomposes), bounded support ($|\text{supp}(P)| \leq k$), log-concave distributions.

1.2 Sequential Regime

When decisions unfold over time with transitions $T : A \times S \to \Delta(S)$ and observations $O : S \to \Delta(\Omega)$, the problem becomes a POMDP. Sequential sufficiency asks whether observation history restricted to I-coordinates determines the optimal policy.

The PSPACEupper bound follows from Papadimitriou & Tsitsiklis (1987). The lower bound uses TQBF reduction: given $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi$, construct n time steps, adversarial transitions at universal quantifiers, agent choices at existential quantifiers, terminal utility φ. Then full observation history is sequentially sufficient iff the QBF is true.

Tractable subcases: fully observable (MDP), bounded horizon, tree-structured transitions, deterministic transitions.

1.3 Substrate Cost

We extend Paper 4's integrity-competence matrix with substrate cost $\kappa(\text{cell}, \text{agentType})$. The matrix verdict is substrate-independent: $\text{verdict}(c)$ is the same for all agent types. But trajectories diverge: $\text{trajectory}(\sigma, \tau_1) \neq \text{trajectory}(\sigma, \tau_2)$ in general. This is the minimal formal hook for substrate in decision theory.

1.4 Temporal Learning

An agent facing sequential problem instances drawn from unknown distribution D can learn the structure class via Bayesian updates. As $k \to \infty$, posterior $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ if structure classes are identifiable. This reduces the abstention frontier: $A_1 \supseteq A_2 \supseteq \ldots$.

1.5 Temporal Integrity

Claims across sequences must satisfy evidence-monotone retractions: an agent retracts a claim only when new evidence contradicts it. Retraction without evidence violates temporal integrity; refinement ($I' \subset I$) is allowed.

1.6 Main Theorems

1. **Theorem 2.6:** STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction.

2. **Theorem 3.6:** SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction.

3. **Propositions 4.1 and 4.2 (Regime Hierarchy):** Static $\subset$ Stochastic $\subset$ Sequential; coNP$\subset$ PP$\subset$ PSPACE.

4. **Theorem 5.2 (Substrate Independence of Verdict):** For all cells c, verdict(c) is κ-independent.

5. **Theorem 5.3 (Substrate Dependence of Trajectory):** There exist τ_1, τ_2, σ with trajectory$(\sigma, \tau_1) \neq$ trajectory(σ, τ_2).

(L: CC58, CC48, CC64, CC63)

1.7 Claim-Typing Bridges (Inherited Static Core)

1. **Typed claim admissibility (inherited static core):** report admissibility is typed by certificate class (ABSTAIN, EXACT, EPSILON(ε)).

2. **Evidence-admissibility equivalence (inherited static core):** first-class evidence-object existence is equivalent to typed report admissibility.

3. **Empty-sufficient characterization (inherited static core):** the $I = \emptyset$ query is exactly universal constancy of the decision boundary, and non-sufficiency is exactly existence of a counterexample pair.

4. **Exact requires evidence (inherited static core):** exact admissibility is equivalent to exact evidence existence.

5. **No evidence implies zero certified confidence (inherited static core):** under signal consistency, no evidence implies zero certified confidence.

[C:P4-core typed-claim chain;R:DC,AR] (L: CC75,IC7,DP2,DP5,IC9,IC17)

1.8 Paper Structure

Section 2: Stochastic decision problems, PP-completeness, tractable cases. Section 3: Sequential (POMDP) problems, PSPACE-completeness, tractable cases. Section 4: Strict hierarchy of regimes. Section 5: Substrate cost formalization and independence results. Section 6: Bayesian structure detection, abstention reduction. Section 7: Temporal integrity constraints. Section 8: Transfer conditions between regimes. Section 9: Related work.

2 Stochastic Decision Problems

We extend the static decision problem to include a probability distribution over states.

2.1 Stochastic Decision Problem

Definition 2.1 (Stochastic Decision Problem)**.** A *stochastic decision problem with coordinate structure* is a tuple $\mathcal{D}_S = (A, X_1, \ldots, X_n, U, P)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *utility function*
- $P \in \Delta(S)$ is a *distribution over states*

Definition 2.2 (Distribution-Conditional Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *stochastically sufficient* for $\mathcal{D}_S$ if:

$$P(s \mid s_I) \otimes U \text{ determines the same optimal action as } P(s) \otimes U.$$

Equivalently, for all $s, s' \in S$ with $s_I = s'_I$:

$$\arg\max_{a \in A} \mathbb{E}_{t \sim P(\cdot \mid s_I)}[U(a,t)] = \arg\max_{a \in A} \mathbb{E}_{t \sim P}[U(a,t)].$$

Definition 2.3 (Expected-Value Sufficiency (Alternative))**.** A coordinate set I is *expected-value sufficient* if:

$$\forall a, a' : \mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I] \implies a \in \mathrm{Opt}_P.$$

where $\mathrm{Opt}_P = \arg\max_{a \in A} \mathbb{E}_{s \sim P}[U(a,s)]$.

Decision 2.4. STOCHASTIC-SUFFICIENCY-CHECK: Given stochastic decision problem $\mathcal{D}_S = (A, S, U, P)$ and coordinate set I, is I stochastically sufficient?

Decision 2.5. STOCHASTIC-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_S$, find the minimal $I \subseteq \{1, \ldots, n\}$ that is stochastically sufficient.

2.2 Complexity: PP-Completeness

Theorem 2.6 (STOCHASTIC-SUFFICIENCY-CHECK is PP-complete)**.** *STOCHASTIC-SUFFICIENCY-CHECK is PP-complete.*

1. **Membership in PP**: Given $\mathcal{D}_S$ and I, we must check whether knowing s_I determines the optimal action. This reduces to comparing expected utilities conditioned on s_I versus unconditioned. For each action pair (a, a'), we compare:

   $$\mathbb{E}_{s \sim P}[U(a,s) \mid s_I] \geq \mathbb{E}_{s \sim P}[U(a',s) \mid s_I].$$

   This requires counting weighted by P, which is in PP.

2. **PP-hardness via MAJSAT reduction**: Given Boolean formula φ on n variables, construct:

   - $S = \{0,1\}^n$ (state space = assignments)
   - P = uniform distribution over S
   - $A = \{\text{accept}, \text{reject}\}$

- $U(\text{accept}, s) = \varphi(s)$, $U(\text{reject}, s) = 1/2$
- $I = \emptyset$

Then $\emptyset$ is stochastically sufficient iff the optimal action is determined without seeing any coordinates:

$$\emptyset \text{ sufficient} \iff \mathbb{E}_{s\sim P}[\varphi(s)] \geq 1/2 \text{ or } < 1/2.$$

This is exactly MAJSAT, which is PP-complete.

(L: CC58, CC55)

Proof. For membership: We reduce to a PPpredicate. For each $a \in A$, compute $\mathbb{E}_P[U(a, s)]$ and compare. The comparison involves summing over $s \in S$ weighted by $P(s)$, which is polynomial in $|S|$ and the representation of P. The decision is whether the argmax is unique or whether knowing s_I changes it—this is a majority-vote comparison, hence in PP.

For hardness: The reduction from MAJSAT constructs the uniform distribution. If φ is true on $\geq 1/2$ of assignments, the expected utility of accepting is $\geq 1/2$, making accept optimal regardless of state observation. If $< 1/2$, reject is optimal. Thus $\emptyset$ is sufficient exactly when MAJSAT is true. ■

Example 2.7 (Stochastic Weather)**.** Consider stochastic umbrella decision:

- $A = \{\text{carry}, \text{don't carry}\}$
- $S = \{\text{rain}, \text{no rain}\}$
- $P(\text{rain}) = 0.3$, $P(\text{no rain}) = 0.7$
- $U(\text{carry}, \text{rain}) = 2$, $U(\text{carry}, \text{no rain}) = -1$
- $U(\text{don't carry}, \text{rain}) = -2$, $U(\text{don't carry}, \text{no rain}) = 0$

Expected utility: $E_P[U(\text{carry})] = 0.3(2)+0.7(-1) = -0.1$, $E_P[U(\text{don't carry})] = 0.3(-2)+0.7(0) = -0.6$. So carry is optimal. If we condition on forecast information, we check whether the conditional distribution changes the argmax.

2.3 Tractable Subcases

1. **Product distributions**: $P = P_1 \otimes \cdots \otimes P_n$. Then

$$\mathbb{E}_{s\sim P}[U(a, s) \mid s_I] = \prod_{i\in I} P_i(s_i) \cdot \mathbb{E}_{s\sim P}[U(a, s) \mid s_I],$$

   so expected utility decomposes. This reduces stochastic sufficiency to static sufficiency per coordinate.

2. **Bounded support**: $|\text{supp}(P)| \leq k$. Enumeration is $O(k \cdot |A|)$, polynomial.

3. **Log-concave distributions**: For ordered state spaces, conditional expectations are monotone and computable via convex optimization.

Claim 2.8 (Tractability under product distributions)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution, then STOCHASTIC-SUFFICIENCY-CHECK reduces to checking static sufficiency on each marginal. *[C:P2.8;R:PD]*

(L: CC60)

Claim 2.9 (Tractability under bounded support)**.** If $|\text{supp}(P)| \leq k$, then STOCHASTIC-SUFFICIENCY-CHECK is solvable in $O(k \cdot |A| \cdot n)$ time. *[C:P2.9;R:BS]*

(L: CC59)

2.4 Bridge from Static

Paper 4 proved that static sufficiency does NOT imply stochastic sufficiency (CC54). A counterexample exists where I is sufficient for (A, S, U) but not for (A, S, U, P).

However, under product distributions, static sufficiency transfers:

Claim 2.10 (Static to stochastic transfer)**.** If $P = P_1 \otimes \cdots \otimes P_n$ is a product distribution and I is statically sufficient for (A, S, U), then I is stochastically sufficient for (A, S, U, P). *[C:P2.10;R:PD]*

(L: CC52)

3 Sequential Decision Problems

We further extend to sequential decision-making with transitions and observations.

3.1 Sequential Decision Problem (POMDP)

Definition 3.1 (Sequential Decision Problem)**.** A *sequential decision problem with coordinate structure* is a tuple $\mathcal{D}_{SEQ} = (A, X_1, \ldots, X_n, U, T, O)$ where:

- A is a finite set of *actions*
- $X_1, \ldots, X_n$ are finite *coordinate spaces*
- $S = X_1 \times \cdots \times X_n$ is the *state space*
- $U : A \times S \to \mathbb{Q}$ is the *stage utility*
- $T : A \times S \to \Delta(S)$ is the *transition kernel*
- $O : S \to \Delta(\Omega)$ is the *observation model*

The agent chooses a policy $\pi : (\Omega^*) \to A$ mapping observation histories to actions. Objective: maximize expected cumulative utility $\mathbb{E}\left[\sum_{t=0}^{T-1} \gamma^t U(a_t, s_t)\right]$ with discount factor $\gamma \in [0, 1)$.

Notation 3.2 (Observation History)**.** Let $\omega_t = (o_0, o_1, \ldots, o_t)$ denote the observation history up to time t. The agent's policy is $\pi(\omega_{t-1}) = a_t$.

Definition 3.3 (Sequential Sufficiency)**.** A coordinate set $I \subseteq \{1, \ldots, n\}$ is *sequentially sufficient* for $\mathcal{D}_{SEQ}$ if the optimal policy based on full observation history is the same as the optimal policy based only on coordinates in I at each time step.

Formally: let π^* be the optimal policy with full observations, and let π_I^* be the optimal policy with observations restricted to I-coordinates. Then $\pi^* = \pi_I^*$ (as functions of observation history).

Decision 3.4. SEQUENTIAL-SUFFICIENCY-CHECK: Given sequential decision problem $\mathcal{D}_{SEQ}$ and coordinate set I, is I sequentially sufficient?

Decision 3.5. SEQUENTIAL-MINIMUM-SUFFICIENT-SET: Given $\mathcal{D}_{SEQ}$, find the minimal I that is sequentially sufficient.

3.2 Complexity: PSPACE-Completeness

Theorem 3.6 (SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete)**.** *SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete.* heorem

1. **Membership in PSPACE**: Optimal POMDP policy computation is in PSPACE(Papadimitriou & Tsitsiklis, 1987). Checking whether a coordinate set I yields the same policy involves comparing value functions, which can be done in polynomial space via alternating polynomial time.

2. **PSPACE-hardness via TQBF reduction**: Given quantified Boolean formula $\forall x_1 \exists x_2 \forall x_3 \ldots \varphi(x_1, \ldots, x_n)$, construct:

   - n time steps, one variable per step
   - At odd steps (universal quantifier): "nature" chooses the variable (adversarial transition)
   - At even steps (existential quantifier): agent chooses the variable (action)
   - Terminal utility: $\varphi(x_1, \ldots, x_n)$
   - Observation: agent sees all previous variables
   - I = full observation history

   Then full observation is sequentially sufficient iff the agent can guarantee $\varphi = \text{true}$ regardless of nature's choices, which holds iff the QBF is true.

(L: CC48, CC47)

Construction 3.7 (TQBF Reduction)**.** Given QBF $\Phi = Q_1 x_1 Q_2 x_2 \ldots Q_n x_n \ \varphi(x_1, \ldots, x_n)$ with alternating quantifiers, construct sequential problem:

- Time steps $t = 1, \ldots, n$
- State at step t encodes $(x_1, \ldots, x_t)$
- At step t: - If $Q_t = \forall$: transition is adversarial (nature chooses both next state and observation) - If $Q_t = \exists$: transition is controlled by agent action (choose x_t)
- Terminal reward: 1 if $\varphi(x_1, \ldots, x_n) = \text{true}$, 0 otherwise
- Full observation reveals all previous variables

Then: full observation is sequentially sufficient $\iff$ there exists a policy guaranteeing $\varphi = \text{true}$ $\iff$ Φ is true.

(L: CC77)

Claim 3.8 (Sequential sufficiency implies static sufficiency)**.** When the horizon $T = 1$ and transitions are deterministic, sequential sufficiency reduces to static sufficiency. *[C:P3.8;R:T=1,DET]*

(L: CC46)

3.3 Tractable Subcases

1. **Fully observable (MDP)**: O is the identity (agent sees full state). Then the problem reduces to a stochastic (or deterministic) problem per time step. Value iteration is polynomial in $|S|, |A|, T$.

2. **Bounded horizon**: $T \leq k$. The state space grows as $O(|S|^T)$, but for fixed T this is polynomial in $|S|$.

3. **Tree-structured transitions**: The transition graph has no cycles. Dynamic programming yields polynomial algorithms.

4. **Deterministic transitions**: $T(s'|a, s)$ is a point mass. The problem reduces to search over deterministic paths.

Claim 3.9 (Tractability under full observability)**.** If O is the identity (fully observable), then SEQUENTIAL-SUFFICIENCY-CHECK reduces to checking stochastic sufficiency at each time step. *[C:P3.9;R:FO]*

(L: CC50)

Claim 3.10 (Tractability under bounded horizon)**.** If $T \leq k$ (constant horizon), then SEQUENTIAL-SUFFICIENCY-CHECK is solvable in time polynomial in $|S|, |A|, k$. *[C:P3.10;R:BH]*

(L: CC49)

3.4 Bridge from Static and Stochastic

Paper 4 proved that static sufficiency does NOT transfer to sequential (horizon > 1): CC18

Similarly, stochastic sufficiency does NOT transfer to sequential in general:

Claim 3.11 (Stochastic to sequential bridge failure)**.** There exists a stochastic sufficient coordinate set I and a sequential problem where I is NOT sequentially sufficient, even when transitions are memoryless. *[C:P3.11;R:CE]*

(L: CC56)

The transfer conditions are:

- Static $\rightarrow$ Sequential: transfers iff $T = 1$ and transitions are deterministic (one-step bridge from Paper 4)

- Stochastic $\rightarrow$ Sequential: transfers iff transitions are memoryless AND observation model is regime-compatible

4 Regime Hierarchy

We establish the strict inclusion hierarchy among decision regimes.

Inclusion	Complexity Classes	Condition
Static $\subset$ Stochastic	coNP$\subset$ PP	Standard assumptions (P $\neq$ coNP)
Stochastic $\subset$ Sequential	PP$\subset$ PSPACE	Standard assumptions (P $\neq$ PP)

Claim 4.1 (Strict inclusion static $\subset$ stochastic)**.** Under standard complexity assumptions (P $\neq$ coNP), there exist stochastic decision problems that are not statically reducible. *[C:P4.1;R:P $\neq$ coNP]*

Claim 4.2 (Strict inclusion stochastic $\subset$ sequential)**.** Under standard complexity assumptions (P $\neq$ PP), there exist sequential decision problems that are not stochastically reducible. *[C:P4.2;R:P $\neq$ PP]*

4.1 Integrity-Resource Bound per Regime

The integrity-resource bound from Paper 4 generalizes to each regime:

1. **Static** (coNP-hard): polynomial-time agent must abstain on some instances
2. **Stochastic** (PP-hard): polynomial-time agent must abstain on MORE instances
3. **Sequential** (PSPACE-hard): polynomial-time agent must abstain on EVEN MORE instances

Formally, if $C_1 \subsetneq C_2$ (complexity classes), then the set of instances where integrity forces abstention under C_1-resources is a strict superset of those under C_2-resources:

$$\{\sigma : \text{integrity forces abstention at } C_1\} \supsetneq \{\sigma : \text{integrity forces abstention at } C_2\}$$

Claim 4.3 (Abstention frontier shifts)**.** As regime complexity increases (static $\to$ stochastic $\to$ sequential), the abstention frontier expands. *[C:P4.3;R:RG]*

(L: CC39)

4.2 Regime Detection

A meta-problem: given a problem instance, which regime is it in?

Decision 4.4. REGIME-DETECTION: Given a problem instance, is it static, stochastic, or sequential?

Conjecture: regime detection is polynomial for the represented family:

- Bounded actions $\to$ static
- Distribution present $\to$ stochastic
- Transitions/observations present $\to$ sequential

(L: CC41)

5 Substrate Cost

We extend the integrity-competence matrix from Paper 4 with substrate cost κ.

5.1 Definition

Definition 5.1 (Substrate Cost Function)**.** $\kappa : (\text{MatrixCell} \times \text{AgentType}) \to (\mathbb{R}_{\geq 0}, \leq)$

Where:

- MatrixCell = (integrity, attempted, competenceAvailable) triple from Paper 4
- AgentType = opaque type representing substrate (silicon, carbon, formal system, etc.)
- The codomain is a totally ordered cost space

Claim 5.2 (Verdict is substrate-independent)**.** For all cells c and agent types τ_1, τ_2:

$$\text{verdict}(c; \tau_1) = \text{verdict}(c; \tau_2).$$

[C:T5.2;R:RG]

(L: CC64)

Claim 5.3 (Trajectory is substrate-dependent)**.** There exist agent types τ_1, τ_2 and problem sequences σ such that:

$$\text{trajectory}(\sigma; \tau_1) \neq \text{trajectory}(\sigma; \tau_2)$$

even though $\text{verdict}(\sigma_t; \tau_1) = \text{verdict}(\sigma_t; \tau_2)$ for all t. *[C:T5.3;R:RG]*

(L: CC63)

5.2 What κ Captures

κ is a placeholder for everything the (A, S, U) tuple doesn't model:

- Memory
- Motivation
- Fatigue
- Identity
- Embodiment

We do NOT decompose κ. We acknowledge it exists, show where it enters the dynamics, and prove that the matrix verdict is κ-independent while the trajectory is κ-dependent.

Claim 5.4 (Minimal formal hook)**.** $\kappa(\text{cell}, \text{agentType})$ is the sufficient statistic of substrate for the trajectory problem. *[C:T5.4;R:RG]*

(L: CC65)

This is the clean separation: the model captures exactly the behaviorally relevant component of substrate, and nothing more.

We do NOT prove that κ captures all of subjective experience. We prove that κ captures all of subjective experience that is *decision-relevant*—i.e., that affects the agent's trajectory through the problem sequence.

5.3 Where Substrate Enters

The matrix verdict is computed from (integrity, attempted, competenceAvailable). None of these depend on substrate. The verdict is universal.

Substrate enters at exactly one point: the COST of being in a cell. This cost determines:

- Whether the agent attempts the next instance or rests
- How the agent allocates resources across a sequence
- The agent's effective learning rate (some substrates learn faster)
- The agent's abstention threshold (some substrates tolerate uncertainty better)

This connects to Paper 5 (Credibility): $\kappa = \kappa_{\text{compute}} + \kappa_{\text{communicate}}$, where $\kappa_{\text{communicate}}$ is bounded by Paper 5's cheap talk framework.

6 Temporal Learning

We formalize how an agent learns structure classes across sequential problem instances.

6.1 Setup

Agent faces a sequence of problem instances $I_1, I_2, \ldots$ drawn from an unknown distribution D over problem structures.

After solving instance I_k (or abstaining), the agent observes:

- Whether I_k belonged to a tractable subcase
- Which structure class C_k was detected (if any)
- The cost paid (substrate-dependent κ)

Claim 6.1 (Bayesian structure detection)**.** Agent maintains posterior $P(C \mid I_1, \ldots, I_k)$ over structure classes. Update rule: standard Bayesian update with likelihood $P(I_k \mid C)$. *[C:T6.1;R:RG]*

(L: CC4)

Convergence: does $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$?

Answer: yes, if the structure classes are identifiable (distinct likelihood functions).

Claim 6.2 (Identifiability implies convergence)**.** If structure classes C are identifiable (distinct likelihood functions), then $P(C \mid I_1, \ldots, I_k) \to \delta_{C*}$ as $k \to \infty$. *[C:P6.2;R:ID]*

(L: CC19)

6.2 PAC Learning Connection

Sample complexity of structure detection: How many instances must the agent solve before it can identify the structure class with probability $\geq 1 - \delta$ and error $\leq \varepsilon$?

Claim 6.3 (Sample complexity conjecture)**.** For the represented family (bounded actions, separable utility, tree structure), sample complexity is polynomial in $1/\varepsilon$ and $\log(1/\delta)$. *[C:J6.3;R:RG]*

This connects to PAC learning theory—the structure classes are the hypothesis space.

6.3 Learning Reduces Abstention

As the agent learns the structure class, it can apply regime-appropriate heuristics. The abstention frontier shrinks over time—not because the problem gets easier, but because the agent's model of the problem gets more accurate.

Formally: let A_k be the set of instances where integrity forces abstention at time k. Then:

$$A_1 \supseteq A_2 \supseteq \ldots \supseteq A_\infty$$

where A_∞ = instances genuinely outside all tractable subcases.

Claim 6.4 (Abstention reduction over time)**.** A_k is decreasing in k: $A_1 \supseteq A_2 \supseteq \ldots$. *[C:T6.4;R:RG]*

(L: CC68)

The agent's trajectory diverges based on κ even though verdicts are κ-independent. Different substrates learn at different rates, have different abstention thresholds, and make different trade-offs between solving and abstaining.

This connects to Paper 5: temporal learning produces costly signals. Each solved instance is a costly signal in Paper 5's sense: producing a correct solution has truth-dependent cost. The agent's track record is costly signal; self-assessment is cheap talk.

7 Temporal Integrity

We formalize consistency of claims across sequences of problem instances.

7.1 Definition

Definition 7.1 (Temporal Integrity)**.** Temporal integrity = consistency of claims across a sequence of instances.

An agent has temporal integrity if:

- Each per-instance claim satisfies Paper 4's integrity constraint
- Retractions are evidence-monotone: the agent retracts a claim only when new evidence contradicts it, never without cause
- Refinements are monotone: if the agent claims I is sufficient at time t, and later claims $I' \subset I$ is sufficient at time $t' > t$, this is a refinement (allowed). If it claims $I' \supset I$, this is a weakening that requires justification.

Decision 7.2. TEMPORAL-INTEGRITY-CHECK: Given a sequence of claims and evidence, is the sequence temporally integrity-preserving?

Claim 7.3 (Retraction with evidence preserves integrity)**.** If an agent claimed I sufficient, observed a counterexample, then retracted, this preserves temporal integrity. *[C:P7.3;R:RG]*

(L: CC70)

Claim 7.4 (Retraction without evidence violates integrity)**.** If an agent claimed I sufficient, observed no new information, then retracted, this violates temporal integrity. *[C:P7.4;R:RG]*

(L: CC71)

Claim 7.5 (Refinement strengthens integrity)**.** If an agent claimed I sufficient, discovers $I' \subset I$ also sufficient, updates claim to I'. This preserves and strengthens temporal integrity. *[C:P7.5;R:RG]*

(L: CC69)

7.2 Complexity of Temporal Integrity

Claim 7.6 (Complexity lower bound)**.** TEMPORAL-INTEGRITY-CHECK is at least as hard as the base regime's SUFFICIENCY-CHECK. *[C:T7.6;R:RG]*

Each claim in the sequence requires verification. The temporal constraint (monotonicity) adds structure but doesn't reduce worst-case complexity.

Paper 4 proves integrity forces abstention. Paper 5 proves claiming integrity is cheap talk. Together: the agent must abstain (Paper 4), cannot credibly communicate why (Paper 5, cheap talk bound), and can only escape by producing costly signals (Paper 5, Theorem 4.1)—which are exactly the mechanized proofs that Paper 4 itself provides.

8 Cross-Regime Transfer

When does a sufficiency result in regime R_1 help in regime R_2?

8.1 Transfer Conditions

1. **Static → Stochastic**: Transfers iff distribution P concentrates on a tractable static subcase. Formally: if $\text{supp}(P)$ lies within a structure-detectable class C, then static sufficiency for C implies stochastic sufficiency under P. Otherwise: bridge failure (already mechanized).

2. **Static → Sequential**: Transfers iff horizon $T = 1$ and transitions are deterministic. This is the one-step bridge (already proved in Paper 4: CC30). For $T > 1$: bridge failure (CC18).

3. **Stochastic → Sequential**: Transfers iff transitions are memoryless AND observation model is regime-compatible. Formally: if $T(s'|a, s)$ depends only on the stochastically sufficient coordinates of s, then stochastic sufficiency lifts to sequential sufficiency. Otherwise: new bridge-failure witness needed.

4. **Sequential → Static**: Always degenerates. A sequential problem with $T = 1$ and deterministic transitions IS a static problem.

Claim 8.1 (Transfer static to stochastic)**.** Static sufficiency transfers to stochastic iff P concentrates on tractable subcase. *[C:T8.1;R:RG]*

(L: CC53)

Claim 8.2 (Transfer stochastic to sequential)**.** Stochastic sufficiency transfers to sequential iff memoryless transitions and regime-compatible observations. *[C:T8.2;R:RG]*

(L: CC57)

8.2 Heuristic Transfer Across Regimes

Paper 4's heuristic reusability bridge generalizes: if structure detection is polynomial in regime R_1, and the detected structure is preserved under regime extension to R_2, then the heuristic transfers.

Key question: when is structure preserved?

- Product distributions preserve separable utility
- Deterministic transitions preserve tree structure

9 Related Work

This paper extends the static decision quotient to stochastic and sequential regimes. Related work falls into several categories.

9.1 Complexity of Dynamic Decision Problems

The complexity of MDPs and POMDPs is well-established: optimal MDP planning is in P, optimal POMDP planning is PSPACE-complete [10, 8, 6]. Our contribution extends this to the meta-question of identifying sufficient coordinates in these settings.

Sufficient statistics [3] is the classical foundation. Our stochastic sufficiency definition connects to this tradition but focuses on decision-relevant information rather than statistical estimation.

9.2 Information and Signaling

Value of information [5, 11] quantifies payment for information. We address a different question: the complexity of identifying which information is sufficient for decision.

Signaling games [12, 9, 7] connect to our temporal integrity and credibility results (Paper 5). The agent must abstain (Paper 4), cannot credibly signal why (Paper 5), and can only escape via costly signals.

9.3 Feature Selection in Machine Learning

Feature selection asks which inputs predict outputs, known NP-hard in general [2, 1, 4]. Our decision-theoretic analog for sufficiency is PP-complete (stochastic) and PSPACE-complete (sequential).

9.4 Regime Complexity Hierarchy

Our hierarchy Static $\subset$ Stochastic $\subset$ Sequential with complexity classes coNP $\subset$ PP $\subset$ PSPACE extends the representation-sensitive complexity results established for MDPs/POMDPs to the meta-level sufficiency problem.

This paper integrates: (i) stochastic and sequential extensions, (ii) substrate cost formalization, (iii) temporal learning and integrity, and (iv) cross-regime transfer conditions.

10 Conclusion

This paper extends the static decision quotient to stochastic and sequential regimes.

Summary of Results

- **Stochastic sufficiency** (Section 2): STOCHASTIC-SUFFICIENCY-CHECK is PP-complete via MAJSAT reduction. Tractable under product distributions, bounded support, log-concave.
- **Sequential sufficiency** (Section 3): SEQUENTIAL-SUFFICIENCY-CHECK is PSPACE-complete via TQBF reduction. Tractable under full observability, bounded horizon, tree structure.
- **Regime hierarchy** (Section 4): Static $\subset$ Stochastic $\subset$ Sequential; coNP $\subset$ PP $\subset$ PSPACE.
- **Substrate cost** (Section 5): Verdict is κ-independent; trajectory is κ-dependent.
- **Temporal learning** (Section 6): Bayesian structure detection reduces abstention over time.
- **Temporal integrity** (Section 7): Evidence-monotone retractions preserve integrity.

These results place stochastic and sequential sufficiency at higher complexity levels than static sufficiency.

Open Questions

- **Exact transfer conditions:** Characterize when stochastic sufficiency transfers to sequential.
- **PAC learning bounds:** Prove sample complexity bounds for structure detection.
- **Temporal integrity complexity:** Tight complexity bounds for TEMPORAL-INTEGRITY-CHECK.

This work extends Paper 4's framework to dynamic settings while preserving the core insight: identifying sufficient information is computationally hard, and integrity forces abstention when resources don't match problem complexity.

11 Lean 4 Proof Listings

The complete Lean 4 formalization extends the Paper 4 artifact. The mechanization consists of 3288 lines across 29 files, with 124 theorem/lemma statements.
Handle IDs. Inline theorem metadata cites compact IDs (for example, `CC4`, `HD2`, `IC3`) instead of full theorem constants.

11.1 Lean Handle ID Map

ID	Full Lean handle
AR1	AccessRegime.le
AR2	AccessRegime.answer_space
AR3	AccessRegime.le_refl
AR4	AccessRegime.le_trans
AR5	AccessRegime.query_space
AU1	AdditiveUtility.isRelevant
AU2	AdditiveUtility.relevantSet

ID	Full Lean handle (continued)
AU3	AdditiveUtility.toProblem
BC1	BooleanCircuit.circuitSize
C1	Complexity.BitStr
C2	Complexity.Decides
C3	Complexity.KarpReduces
C4	Complexity.Lang
C5	Complexity.ManyOneReducesPoly
C6	Complexity.P
C7	Complexity.PolyTime
C8	Complexity.bitEnc
C9	Counted.bind
C10	Counted.bind_steps
C11	Counted.pure
C12	Counted.pure_steps
C13	Counted.result
C14	Counted.steps
C15	Counted.steps_eq_fst
C16	Counted.tick
C17	Counted.tick_bind_steps
C31	Clause3.eval
CC1	DecisionQuotient.ClaimClosure.RegimeSimulation
CC2	DecisionQuotient.ClaimClosure.adq_ordering
CC3	DecisionQuotient.ClaimClosure.agent_transfer_licensed_iff_snapshot
CC4	DecisionQuotient.ClaimClosure.anchor_sigma2p_complete_conditional
CC5	DecisionQuotient.ClaimClosure.anchor_sigma2p_reduction_core
CC6	DecisionQuotient.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset
CC7	DecisionQuotient.ClaimClosure.bounded_actions_detectable
CC8	DecisionQuotient.ClaimClosure.bridge_boundary_represented_family
CC9	DecisionQuotient.ClaimClosure.bridge_failure_witness_non_one_step
CC10	DecisionQuotient.ClaimClosure.bridge_transfer_iff_one_step_class
CC11	DecisionQuotient.ClaimClosure.certified_total_bits_split_core
CC12	DecisionQuotient.ClaimClosure.cost_asymmetry_eth_conditional
CC13	DecisionQuotient.ClaimClosure.declaredBudgetSlice
CC14	DecisionQuotient.ClaimClosure.declaredRegimeFamily_complete
CC15	DecisionQuotient.ClaimClosure.declared_physics_no_universal_exact_certifier_core
CC16	DecisionQuotient.ClaimClosure.dichotomy_conditional
CC17	DecisionQuotient.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core
CC18	DecisionQuotient.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
CC19	DecisionQuotient.ClaimClosure.exact_certainty_inflation_under_hardness_core
CC20	DecisionQuotient.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core
CC21	DecisionQuotient.ClaimClosure.exact_raw_only_of_no_exact_admissible_core
CC22	DecisionQuotient.ClaimClosure.explicit_assumptions_required_of_not_excused_core
CC23	DecisionQuotient.ClaimClosure.explicit_state_upper_core
CC24	DecisionQuotient.ClaimClosure.hard_family_all_coords_core
CC25	DecisionQuotient.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
CC26	DecisionQuotient.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
CC27	DecisionQuotient.ClaimClosure.information_barrier_opt_oracle_core
CC28	DecisionQuotient.ClaimClosure.information_barrier_state_batch_core
CC29	DecisionQuotient.ClaimClosure.information_barrier_value_entry_core
CC30	DecisionQuotient.ClaimClosure.integrity_resource_bound_for_sufficiency
CC31	DecisionQuotient.ClaimClosure.integrity_universal_applicability_core
CC32	DecisionQuotient.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
CC33	DecisionQuotient.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice
CC34	ComplexityClass.lt
CC35	DecisionQuotient.ClaimClosure.minsuff_collapse_to_conp_conditional

ID	Full Lean handle (continued)
CC36	DecisionQuotient.ClaimClosure.minsuff_conp_complete_conditional
CC37	DecisionQuotient.ClaimClosure.no_auto_minimize_of_p_neq_conp
CC38	DecisionQuotient.ClaimClosure.no_exact_claim_admissible_under_hardness_core
CC39	DecisionQuotient.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
CC40	DecisionQuotient.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized
CC41	DecisionQuotient.ClaimClosure.no_uncertified_exact_claim_core
CC42	DecisionQuotient.ClaimClosure.one_step_bridge
CC43	DecisionQuotient.ClaimClosure.oracle_lattice_transfer_as_regime_simulation
CC44	DecisionQuotient.ClaimClosure.physical_crossover_above_cap_core
CC45	DecisionQuotient.ClaimClosure.physical_crossover_core
CC46	DecisionQuotient.ClaimClosure.physical_crossover_hardness_core
CC47	DecisionQuotient.ClaimClosure.physical_crossover_policy_core
CC48	DecisionQuotient.ClaimClosure.process_bridge_failure_witness
CC49	DecisionQuotient.ClaimClosure.query_obstruction_boolean_corollary
CC50	DecisionQuotient.ClaimClosure.query_obstruction_finite_state_core
CC51	DecisionQuotient.ClaimClosure.regime_core_claim_proved
CC52	DecisionQuotient.ClaimClosure.regime_simulation_transfers_hardness
CC53	DecisionQuotient.ClaimClosure.reusable_heuristic_of_detectable
CC54	DecisionQuotient.ClaimClosure.selectorSufficient_not_implies_setSufficient
CC55	DecisionQuotient.ClaimClosure.separable_detectable
CC56	DecisionQuotient.ClaimClosure.snapshot_vs_process_typed_boundary
CC57	DecisionQuotient.ClaimClosure.standard_assumption_ledger_unpack
CC58	DecisionQuotient.ClaimClosure.stochastic_objective_bridge_can_fail_on_sufficiency
CC59	DecisionQuotient.ClaimClosure.subproblem_hardness_lifts_to_full
CC60	DecisionQuotient.ClaimClosure.subproblem_transfer_as_regime_simulation
CC61	DecisionQuotient.ClaimClosure.sufficiency_conp_complete_conditional
CC62	DecisionQuotient.ClaimClosure.sufficiency_conp_reduction_core
CC63	DecisionQuotient.ClaimClosure.sufficiency_iff_dq_ratio
CC64	DecisionQuotient.ClaimClosure.sufficiency_iff_projectedOptCover_eq_opt
CC65	DecisionQuotient.ClaimClosure.thermo_conservation_additive_core
CC66	DecisionQuotient.ClaimClosure.thermo_energy_carbon_lift_core
CC67	DecisionQuotient.ClaimClosure.thermo_eventual_lift_core
CC68	DecisionQuotient.ClaimClosure.thermo_hardness_bundle_core
CC69	DecisionQuotient.ClaimClosure.thermo_mandatory_cost_core
CC70	DecisionQuotient.ClaimClosure.tractable_bounded_core
CC71	DecisionQuotient.ClaimClosure.tractable_separable_core
CC72	DecisionQuotient.ClaimClosure.tractable_subcases_conditional
CC73	DecisionQuotient.ClaimClosure.tractable_tree_core
CC74	DecisionQuotient.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency
CC75	DecisionQuotient.ClaimClosure.tree_structure_detectable
CC76	DecisionQuotient.ClaimClosure.typed_claim_admissibility_core
CC77	DecisionQuotient.ClaimClosure.typed_static_class_completeness
CC78	DecisionQuotient.ClaimClosure.universal_solver_framing_core
CDP1	ComputableDecisionProblem.checkSufficiency
CDP2	ComputableDecisionProblem.checkSufficiency_iff_abstract
CDP3	ComputableDecisionProblem.computeOpt
CDP4	ComputableDecisionProblem.computeOpt_eq_abstract
CDP5	ComputableDecisionProblem.decisionEquivDec
CDP6	ComputableDecisionProblem.decisionEquivDec_iff
CDP7	ComputableDecisionProblem.decisionEquivDec_iff_abstract
CDP8	ComputableDecisionProblem.insufficiencyWitnesses
CDP9	ComputableDecisionProblem.isOptimalDec
CDP10	ComputableDecisionProblem.isOptimalDec_iff
CDP11	ComputableDecisionProblem.isOptimalDec_iff_abstract

ID	Full Lean handle (continued)
CDP12	ComputableDecisionProblem.mem_computeOpt_iff
CDP13	ComputableDecisionProblem.toAbstract
CDP14	ComputableDecisionProblem.verifyWitness
CR1	DecisionQuotient.ConfigReduction.config_sufficiency_iff_behavior_preserving
DP1	DecisionQuotient.DecisionProblem.minimalSufficient_iff_relevant
DP2	DecisionQuotient.DecisionProblem.relevantSet_is_minimal
DP3	DecisionQuotient.DecisionProblem.sufficient_implies_selectorSufficient
DP4	DecisionQuotient.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
DP5	DecisionQuotient.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient
DP6	DecisionProblem.anchorSufficient
DP7	DecisionProblem.classMonotoneOn
DP8	DecisionProblem.constant_opt_all_sufficient
DP9	DecisionProblem.DecisionEquiv
DP10	DecisionProblem.DecisionQuotientType
DP11	DecisionProblem.Opt
DP12	DecisionProblem.OptQuotient
DP13	DecisionProblem.SelectedAction
DP14	DecisionProblem.decisionEquiv_refl
DP15	DecisionProblem.decisionEquiv_symm
DP16	DecisionProblem.decisionEquiv_trans
DP17	DecisionProblem.decisionSetoid
DP18	DecisionProblem.dominant_all_sufficient
DP19	DecisionProblem.dominant_unique
DP20	DecisionProblem.edgeOnComplement
DP21	DecisionProblem.edgeOnComplement_iff_not_sufficient
DP22	DecisionProblem.emptySet_not_sufficient_iff_exists_opt_difference
DP23	DecisionProblem.emptySet_sufficient_iff_constant
DP24	DecisionProblem.epsOpt
DP25	DecisionProblem.epsOpt_zero_eq_opt
DP26	DecisionProblem.erase_of_not_mem
DP27	DecisionProblem.factors_implies_respects
DP28	DecisionProblem.hasConstantOpt
DP29	DecisionProblem.hasConstantOpt'
DP30	DecisionProblem.hasDominant
DP31	DecisionProblem.irrelevant_iff_not_relevant
DP32	DecisionProblem.isEpsilonSufficient
DP33	DecisionProblem.isIrrelevant
DP34	DecisionProblem.isMinimalSufficient
DP35	DecisionProblem.isMinimalSufficient'
DP36	DecisionProblem.isOptimal
DP37	DecisionProblem.isRelevant
DP38	DecisionProblem.isSelectorSufficient
DP39	DecisionProblem.isSufficient
DP40	DecisionProblem.isSufficientAt
DP41	DecisionProblem.isSufficientOnSet
DP42	DecisionProblem.minimalSufficient_all_relevant'
DP43	DecisionProblem.minimalSufficient_eq_relevant'
DP44	DecisionProblem.minimalSufficient_iff_relevant
DP45	DecisionProblem.monotoneIn
DP46	DecisionProblem.nonnegativelyMonotoneCoord
DP47	DecisionProblem.not_sufficient_iff_exists_counterexample
DP48	DecisionProblem.numOptClasses
DP49	DecisionProblem.numOptClasses_le
DP50	DecisionProblem.numOptClasses_pos
DP51	DecisionProblem.opt_eq_optQuotient_comp

ID	Full Lean handle (continued)
DP52	DecisionProblem.opt_factors_through_quotient
DP53	DecisionProblem.preservesOpt
DP54	DecisionProblem.preservesOptStrong
DP55	DecisionProblem.quotientEntropy
DP56	DecisionProblem.quotientEntropy_nonneg
DP57	DecisionProblem.quotientEntropy_zero_of_constant
DP58	DecisionProblem.quotientMap
DP59	DecisionProblem.quotientMap_preservesOpt
DP60	DecisionProblem.quotientSize
DP61	DecisionProblem.quotient_is_coarsest
DP62	DecisionProblem.quotient_represents_opt_equiv
DP63	DecisionProblem.relevantSet
DP64	DecisionProblem.relevantSet_is_minimal
DP65	DecisionProblem.relevant_necessary
DP66	DecisionProblem.sufficientSets
DP67	DecisionProblem.sufficientSets_principal'
DP68	DecisionProblem.sufficientSets_upward
DP69	DecisionProblem.sufficientSets_upward_closed
DP70	DecisionProblem.sufficient_contains_all_relevant
DP71	DecisionProblem.sufficient_contains_relevant
DP72	DecisionProblem.sufficient_erase_irrelevant'
DP73	DecisionProblem.sufficient_iff_zeroEpsilonSufficient
DP74	DecisionProblem.sufficient_implies_selectorSufficient
DP75	DecisionProblem.sufficient_superset
DP76	DecisionProblem.univ_sufficient_of_injective
DQ1	DecisionQuotient.AdditiveUtility
DQ2	DecisionQuotient.AssignX
DQ3	DecisionQuotient.AssignY
DQ4	DecisionQuotient.Assignment
DQ5	DecisionQuotient.BinaryState
DQ6	DecisionQuotient.BooleanCircuit
DQ7	DecisionQuotient.CircuitDecisionProblem
DQ8	DecisionQuotient.ClaimClosure.agreeOn_refl
DQ9	DecisionQuotient.ClaimClosure.agreeOn_symm
DQ10	DecisionQuotient.ClaimClosure.agreeOn_trans
DQ11	DecisionQuotient.ClaimClosure.irrelevantOn_implies_not_relevantOn
DQ12	DecisionQuotient.ClaimClosure.isIrrelevantOn
DQ13	DecisionQuotient.ClaimClosure.isRelevantOn
DQ14	DecisionQuotient.Clause3
DQ15	DecisionQuotient.ComputableDecisionProblem
DQ16	DecisionQuotient.ConfigReduction.ConfigAction
DQ17	DecisionQuotient.ConfigReduction.behaviorPreserving
DQ18	DecisionQuotient.ConfigReduction.configDecisionProblem
DQ19	DecisionQuotient.ConfigReduction.configUtility
DQ20	DecisionQuotient.ConfigReduction.configUtility_le_one
DQ21	DecisionQuotient.ConfigReduction.no_occurs_iff_of_behaviorPreserving
DQ22	DecisionQuotient.ConfigReduction.none_mem_Opt_iff_no_occurs
DQ23	DecisionQuotient.ConfigReduction.some_mem_Opt_iff_occurs
DQ24	DecisionQuotient.ConfigReduction.some_mem_Opt_of_occurs
DQ25	DecisionQuotient.ConfigReduction.some_not_mem_Opt_of_not_occurs
DQ26	DecisionQuotient.CoordinateSpace
DQ27	DecisionQuotient.Counted
DQ28	DecisionQuotient.DQAction
DQ29	DecisionQuotient.DQInstance
DQ30	DecisionQuotient.DecisionProblem

ID	Full Lean handle (continued)
DQ31	DecisionQuotient.EncodedDecisionProblem
DQ32	DecisionQuotient.ExistsForallFormula
DQ33	DecisionQuotient.ExistsForallFormula.embedVar
DQ34	DecisionQuotient.ExistsForallFormula.eval_padUniversal
DQ35	DecisionQuotient.ExistsForallFormula.padUniversal
DQ36	DecisionQuotient.ExistsForallFormula.restrictY
DQ37	DecisionQuotient.ExistsForallFormula.satisfiable
DQ38	DecisionQuotient.ExistsForallFormula.satisfiable_iff_padUniversal
DQ39	DecisionQuotient.ExistsForallFormula.satisfiedBy
DQ40	DecisionQuotient.ExistsForallFormula.sumAssignment
DQ41	DecisionQuotient.ExistsForallFormula.sumAssignment_embed
DQ42	DecisionQuotient.FiniteDecisionProblem
DQ43	DecisionQuotient.FiniteDecisionProblem.decisionQuotient
DQ44	DecisionQuotient.FiniteDecisionProblem.isSufficient
DQ45	DecisionQuotient.FiniteDecisionProblem.isSufficient_superset
DQ46	DecisionQuotient.FiniteDecisionProblem.mem_optimalActions_iff
DQ47	DecisionQuotient.FiniteDecisionProblem.optimalActions
DQ48	DecisionQuotient.FiniteDecisionProblem.optimalActions_subset_actions
DQ49	DecisionQuotient.FiniteDecisionProblem.toDecisionProblem
DQ50	DecisionQuotient.HardnessDistribution.IsEventuallyConstant
DQ51	DecisionQuotient.HardnessDistribution.SolutionArchitecture
DQ52	DecisionQuotient.HardnessDistribution.SpecificationProblem
DQ53	DecisionQuotient.HardnessDistribution.amortizationThreshold
DQ54	DecisionQuotient.HardnessDistribution.amortization_threshold_native_manual
DQ55	DecisionQuotient.HardnessDistribution.centralization_step_reduces_total
DQ56	DecisionQuotient.HardnessDistribution.centralized_constant
DQ57	DecisionQuotient.HardnessDistribution.centralized_minimal_errors
DQ58	DecisionQuotient.HardnessDistribution.distributed_errors_grow
DQ59	DecisionQuotient.HardnessDistribution.distributed_multiplies
DQ60	DecisionQuotient.HardnessDistribution.dof_conservation
DQ61	DecisionQuotient.HardnessDistribution.errorSites
DQ62	DecisionQuotient.HardnessDistribution.gapCard
DQ63	DecisionQuotient.HardnessDistribution.generalizedTotalDOF
DQ64	DecisionQuotient.HardnessDistribution.generalized_right_dominates_wrong_pointwise
DQ65	DecisionQuotient.HardnessDistribution.hardnessEfficiency
DQ66	DecisionQuotient.HardnessDistribution.less_distributed_less_total
DQ67	DecisionQuotient.HardnessDistribution.leverageRatio
DQ68	DecisionQuotient.HardnessDistribution.linear_lt_exponential_eventually
DQ69	DecisionQuotient.HardnessDistribution.linear_represents_saturating_only_zero_slope
DQ70	DecisionQuotient.HardnessDistribution.manualApproach
DQ71	DecisionQuotient.HardnessDistribution.manual_errors_grow
DQ72	DecisionQuotient.HardnessDistribution.manual_is_wrong
DQ73	DecisionQuotient.HardnessDistribution.nativeTypeSystem
DQ74	DecisionQuotient.HardnessDistribution.native_is_right
DQ75	DecisionQuotient.HardnessDistribution.native_minimal_errors
DQ76	DecisionQuotient.HardnessDistribution.right_fewer_error_sites
DQ77	DecisionQuotient.HardnessDistribution.saturatingSiteCost
DQ78	DecisionQuotient.HardnessDistribution.simplicityTax
DQ79	DecisionQuotient.HardnessDistribution.simplicityTax_conservation
DQ80	DecisionQuotient.HardnessDistribution.totalDOF
DQ81	DecisionQuotient.HardnessDistribution.totalDOF_eventually_constant_of_zero_distributed
DQ82	DecisionQuotient.HardnessDistribution.totalExternalWork
DQ83	DecisionQuotient.HardnessDistribution.totalExternalWork_eq_n_mul_gapCard
DQ84	DecisionQuotient.HardnessDistribution.zero_distributed_of_totalDOF_eventually_constant
DQ85	DecisionQuotient.InP

ID	Full Lean handle (continued)
DQ86	DecisionQuotient.InsufficiencyWitness
DQ87	DecisionQuotient.IntegrityCompetence.CertifyingSolver
DQ88	DecisionQuotient.IntegrityCompetence.ClaimAdmissible
DQ89	DecisionQuotient.IntegrityCompetence.CompetentOn
DQ90	DecisionQuotient.IntegrityCompetence.EpsilonCompetentOn
DQ91	DecisionQuotient.IntegrityCompetence.EpsilonRelation
DQ92	DecisionQuotient.IntegrityCompetence.Regime
DQ93	DecisionQuotient.IntegrityCompetence.SolverIntegrity
DQ94	DecisionQuotient.IntegrityCompetence.admissibleIrrationalCount
DQ95	DecisionQuotient.IntegrityCompetence.admissibleRationalCount
DQ96	DecisionQuotient.IntegrityCompetence.admissible_epsilon_implies_integrity
DQ97	DecisionQuotient.IntegrityCompetence.admissible_exact_implies_integrity
DQ98	DecisionQuotient.IntegrityCompetence.alwaysAbstain
DQ99	DecisionQuotient.IntegrityCompetence.alwaysAbstain_integrity
DQ100	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_eq_raw_plus_overhead
DQ101	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_gt_raw_of_evidence
DQ102	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_evidence
DQ103	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_of_no_evidence
DQ104	DecisionQuotient.IntegrityCompetence.claim_admissible_abstain
DQ105	DecisionQuotient.IntegrityCompetence.claim_admissible_epsilon_iff
DQ106	DecisionQuotient.IntegrityCompetence.claim_admissible_exact_iff
DQ107	DecisionQuotient.IntegrityCompetence.claim_admissible_of_evidence
DQ108	DecisionQuotient.IntegrityCompetence.competence_implies_integrity
DQ109	DecisionQuotient.IntegrityCompetence.competent_has_coverage
DQ110	DecisionQuotient.IntegrityCompetence.completion_fraction_defined_of_declared_bound
DQ111	DecisionQuotient.IntegrityCompetence.epsilon_admissible_iff_raw_lt_certifiedTotal
DQ112	DecisionQuotient.IntegrityCompetence.epsilon_competence_implies_integrity
DQ113	DecisionQuotient.IntegrityCompetence.evidence_nonempty_iff_claim_admissible
DQ114	DecisionQuotient.IntegrityCompetence.evidence_of_claim_admissible
DQ115	DecisionQuotient.IntegrityCompetence.exactCertaintyInflation_iff_no_exact_competence
DQ116	DecisionQuotient.IntegrityCompetence.exact_admissible_iff_raw_lt_certifiedTotal
DQ117	DecisionQuotient.IntegrityCompetence.exact_claim_admissible_iff_exact_evidence_nonempty
DQ118	DecisionQuotient.IntegrityCompetence.exact_claim_requires_evidence
DQ119	DecisionQuotient.IntegrityCompetence.exact_raw_eq_certifiedTotal_iff_ exactCertaintyInflation
DQ120	DecisionQuotient.IntegrityCompetence.exact_raw_only_of_no_exact_admissible
DQ121	DecisionQuotient.IntegrityCompetence.inducedRelation
DQ122	DecisionQuotient.IntegrityCompetence.integrity_forces_abstention
DQ123	DecisionQuotient.IntegrityCompetence.integrity_not_competent_of_nonempty_scope
DQ124	DecisionQuotient.IntegrityCompetence.integrity_resource_bound
DQ125	DecisionQuotient.IntegrityCompetence.no_completion_fraction_without_declared_bound
DQ126	DecisionQuotient.IntegrityCompetence.no_uncertified_epsilon_claim
DQ127	DecisionQuotient.IntegrityCompetence.no_uncertified_exact_claim
DQ128	DecisionQuotient.IntegrityCompetence.overModelVerdict
DQ129	DecisionQuotient.IntegrityCompetence.overModelVerdict_inadmissible_iff
DQ130	DecisionQuotient.IntegrityCompetence.overModelVerdict_irrational_iff
DQ131	DecisionQuotient.IntegrityCompetence.overModelVerdict_rational_iff
DQ132	DecisionQuotient.IntegrityCompetence.percentZero
DQ133	DecisionQuotient.IntegrityCompetence.program_framed_as_solver
DQ134	DecisionQuotient.IntegrityCompetence.report_admissible_implies_raw_lt_certifiedTotal
DQ135	DecisionQuotient.IntegrityCompetence.report_not_admissible_implies_raw_eq_certifiedTotal
DQ136	DecisionQuotient.IntegrityCompetence.report_raw_eq_certifiedTotal_iff_certaintyInflation
DQ137	DecisionQuotient.IntegrityCompetence.rlffBaseReward
DQ138	DecisionQuotient.IntegrityCompetence.rlffReward
DQ139	DecisionQuotient.IntegrityCompetence.rlffReward_abstain

ID	Full Lean handle (continued)
DQ140	DecisionQuotient.IntegrityCompetence.rlffReward_of_evidence
DQ141	DecisionQuotient.IntegrityCompetence.rlffReward_of_no_evidence
DQ142	DecisionQuotient.IntegrityCompetence.rlff_abstain_strictly_prefers_no_certificates
DQ143	DecisionQuotient.IntegrityCompetence.rlff_maximizer_has_evidence
DQ144	DecisionQuotient.IntegrityCompetence.rlff_maximizer_is_admissible
DQ145	DecisionQuotient.IntegrityCompetence.self_reflected_confidence_not_certification
DQ146	DecisionQuotient.IntegrityCompetence.signal_certified_positive_implies_admissible
DQ147	DecisionQuotient.IntegrityCompetence.signal_consistent_of_claim_admissible
DQ148	DecisionQuotient.IntegrityCompetence.signal_exact_no_competence_forces_zero_certified
DQ149	DecisionQuotient.IntegrityCompetence.signal_no_evidence_forces_zero_certified
DQ150	DecisionQuotient.IntegrityCompetence.solverIntegrity_substrate_parametric
DQ151	DecisionQuotient.IntegrityCompetence.solverOfPartialMap
DQ152	DecisionQuotient.IntegrityCompetence.solverOfPartialMap_integrity
DQ153	DecisionQuotient.IntegrityCompetence.steps_run_scalar_always_defined
DQ154	DecisionQuotient.IntegrityCompetence.steps_run_scalar_falsifiable
DQ155	DecisionQuotient.IntegrityCompetence.zero_epsilon_competence_iff_exact
DQ156	DecisionQuotient.InteriorVerification.CoordScore
DQ157	DecisionQuotient.InteriorVerification.GoalClass
DQ158	DecisionQuotient.InteriorVerification.InteriorDominanceVerifiable
DQ159	DecisionQuotient.InteriorVerification.TautologicalSetIdentifiable
DQ160	DecisionQuotient.InteriorVerification.agreeOnSet
DQ161	DecisionQuotient.InteriorVerification.interiorParetoDominates
DQ162	DecisionQuotient.InteriorVerification.interior_certificate_implies_non_rejection
DQ163	DecisionQuotient.InteriorVerification.interior_dominance_implies_universal_non_rejection
DQ164	DecisionQuotient.InteriorVerification.interior_dominance_not_full_sufficiency
DQ165	DecisionQuotient.InteriorVerification.interior_verification_tractable_certificate
DQ166	DecisionQuotient.InteriorVerification.not_sufficientOnSet_of_counterexample
DQ167	DecisionQuotient.InteriorVerification.singleton_coordinate_interior_certificate
DQ168	DecisionQuotient.IsPolynomialTime
DQ169	DecisionQuotient.Literal
DQ170	DecisionQuotient.MinSufficientSetInstance
DQ171	DecisionQuotient.Outside
DQ172	DecisionQuotient.PhysicalBudgetCrossover.EncodingSizeModel
DQ173	DecisionQuotient.PhysicalBudgetCrossover.ExplicitInfeasible
DQ174	DecisionQuotient.PhysicalBudgetCrossover.ExplicitUnbounded
DQ175	DecisionQuotient.PhysicalBudgetCrossover.HasCrossover
DQ176	DecisionQuotient.PhysicalBudgetCrossover.SuccinctBoundedBy
DQ177	DecisionQuotient.PhysicalBudgetCrossover.SuccinctFeasible
DQ178	DecisionQuotient.PhysicalBudgetCrossover.crossover_for_all_budgets_above_cap
DQ179	DecisionQuotient.PhysicalBudgetCrossover.crossover_hardness_bundle
DQ180	DecisionQuotient.PhysicalBudgetCrossover.crossover_integrity_policy
DQ181	DecisionQuotient.PhysicalBudgetCrossover.has_crossover_of_bounded_succinct_unbounded_explicit
DQ182	DecisionQuotient.PhysicalBudgetCrossover.has_crossover_of_witness
DQ183	DecisionQuotient.Physics.Instantiation.Configuration
DQ184	DecisionQuotient.Physics.Instantiation.DecisionCircuit
DQ185	DecisionQuotient.Physics.Instantiation.DecisionInterpretation
DQ186	DecisionQuotient.Physics.Instantiation.EnergyLandscape
DQ187	DecisionQuotient.Physics.Instantiation.LandauerBound
DQ188	DecisionQuotient.Physics.Instantiation.Molecule
DQ189	DecisionQuotient.Physics.Instantiation.MoleculeAsCircuit
DQ190	DecisionQuotient.Physics.Instantiation.MoleculeAsDecisionCircuit
DQ192	DecisionQuotient.Physics.Instantiation.Reaction
DQ193	DecisionQuotient.Physics.Instantiation.k_Boltzmann
DQ194	DecisionQuotient.PolyReduction

ID	Full Lean handle (continued)
DQ195	DecisionQuotient.PolyTimeApprox
DQ196	DecisionQuotient.PolytimeElicitationMechanism
DQ197	DecisionQuotient.Prior
DQ198	DecisionQuotient.ProductSpace
DQ199	DecisionQuotient.QBFState
DQ200	DecisionQuotient.QueryAlgorithm
DQ201	DecisionQuotient.QueryTranscript
DQ202	DecisionQuotient.SAT3Instance
DQ203	DecisionQuotient.SAT3Instance.eval
DQ204	DecisionQuotient.SAT3Instance.eval_eq_true_iff
DQ205	DecisionQuotient.SAT3Instance.satisfiable
DQ206	DecisionQuotient.SAT3Instance.satisfiedBy
DQ207	DecisionQuotient.SeparableUtility
DQ208	DecisionQuotient.SetComparisonInstance
DQ209	DecisionQuotient.SharpSATInstance
DQ210	DecisionQuotient.Sigma2PHardness.GadgetState
DQ211	DecisionQuotient.Sigma2PHardness.I_eq_I_of_encodeX
DQ212	DecisionQuotient.Sigma2PHardness.I_of_encodeX_subset
DQ213	DecisionQuotient.Sigma2PHardness.I_of_x
DQ214	DecisionQuotient.Sigma2PHardness.I_of_x_card
DQ215	DecisionQuotient.Sigma2PHardness.bit
DQ216	DecisionQuotient.Sigma2PHardness.bit_le_one
DQ217	DecisionQuotient.Sigma2PHardness.bit_lt_two
DQ218	DecisionQuotient.Sigma2PHardness.cCoord
DQ219	DecisionQuotient.Sigma2PHardness.cCoord_inj
DQ220	DecisionQuotient.Sigma2PHardness.cCoord_inj_bits
DQ221	DecisionQuotient.Sigma2PHardness.cCoord_injective
DQ222	DecisionQuotient.Sigma2PHardness.cCoord_val
DQ223	DecisionQuotient.Sigma2PHardness.cCoord_val_lt_of_lt
DQ224	DecisionQuotient.Sigma2PHardness.disjoint_I_of_x00_x11
DQ225	DecisionQuotient.Sigma2PHardness.encodeX
DQ226	DecisionQuotient.Sigma2PHardness.encodeX_I_of_x
DQ227	DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant
DQ228	DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant_iff_mem_relevant
DQ229	DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
DQ230	DecisionQuotient.Sigma2PHardness.exactlyIdentifiesRelevant_implies_representationGap_zero
DQ231	DecisionQuotient.Sigma2PHardness.goodEq
DQ232	DecisionQuotient.Sigma2PHardness.goodEq_x00
DQ233	DecisionQuotient.Sigma2PHardness.goodEq_x11
DQ234	DecisionQuotient.Sigma2PHardness.mem_I_of_x_iff
DQ235	DecisionQuotient.Sigma2PHardness.mem_relevantFinset_iff
DQ236	DecisionQuotient.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
DQ237	DecisionQuotient.Sigma2PHardness.min_sufficient_set_iff_relevant_card
DQ238	DecisionQuotient.Sigma2PHardness.not_goodEq_x01
DQ239	DecisionQuotient.Sigma2PHardness.not_sufficient_of_missing_relevantFinset
DQ240	DecisionQuotient.Sigma2PHardness.relevantFinset
DQ241	DecisionQuotient.Sigma2PHardness.representationGap
DQ242	DecisionQuotient.Sigma2PHardness.representationGap_eq_waste_plus_missing
DQ243	DecisionQuotient.Sigma2PHardness.representationGap_eq_zero_iff
DQ244	DecisionQuotient.Sigma2PHardness.representationGap_missing_eq_gapCard
DQ245	DecisionQuotient.Sigma2PHardness.representationGap_zero_iff_exactlyIdentifiesRelevant
DQ246	DecisionQuotient.Sigma2PHardness.representationGap_zero_iff_minimalSufficient
DQ247	DecisionQuotient.Sigma2PHardness.representationGap_zero_iff_sufficient_and_subset_relevantFinset

ID	Full Lean handle (continued)
DQ248	DecisionQuotient.Sigma2PHardness.representationGap_zero_implies_sufficient
DQ249	DecisionQuotient.Sigma2PHardness.sufficient_iff_relevantFinset_subset
DQ250	DecisionQuotient.Sigma2PHardness.sufficient_iff_relevant_subset
DQ251	DecisionQuotient.Sigma2PHardness.sufficient_of_contains_relevant
DQ252	DecisionQuotient.Sigma2PHardness.two_mul_add_one_lt_two_mul
DQ253	DecisionQuotient.Sigma2PHardness.univ_sufficient_bool
DQ254	DecisionQuotient.Sigma2PHardness.validI
DQ255	DecisionQuotient.Sigma2PHardness.validI_I_of_x
DQ256	DecisionQuotient.Sigma2PHardness.validI_iff_exists_x
DQ257	DecisionQuotient.Sigma2PHardness.vector1_obstruction
DQ258	DecisionQuotient.Sigma2PHardness.x00
DQ259	DecisionQuotient.Sigma2PHardness.x01
DQ260	DecisionQuotient.Sigma2PHardness.x11
DQ261	DecisionQuotient.Sigma2PHardness.yCoord
DQ262	DecisionQuotient.Sigma2PHardness.yCoord_val
DQ263	DecisionQuotient.Signal
DQ264	DecisionQuotient.StateAbstraction
DQ265	DecisionQuotient.StateBatchQuery
DQ266	DecisionQuotient.StructuredUtility
DQ267	DecisionQuotient.SufficiencyCheckInstance
DQ268	DecisionQuotient.SufficiencyInstance
DQ269	DecisionQuotient.Summary.bounded_actions_tractable
DQ270	DecisionQuotient.Summary.complexity_dichotomy
DQ271	DecisionQuotient.Summary.conp_completeness
DQ272	DecisionQuotient.Summary.eth_lower_bound
DQ273	DecisionQuotient.Summary.min_sufficient_inapproximability
DQ274	DecisionQuotient.Summary.parameterized_results
DQ275	DecisionQuotient.Summary.separable_utility_tractable
DQ276	DecisionQuotient.Summary.sharp_p_hardness
DQ277	DecisionQuotient.Summary.tractability_tightness
DQ278	DecisionQuotient.Summary.tree_structure_tractable
DQ279	DecisionQuotient.ThermodynamicLift.ThermoModel
DQ280	DecisionQuotient.ThermodynamicLift.carbonLowerBound
DQ281	DecisionQuotient.ThermodynamicLift.carbon_lower_additive
DQ282	DecisionQuotient.ThermodynamicLift.carbon_lower_from_bits_lower
DQ283	DecisionQuotient.ThermodynamicLift.carbon_lower_mandatory
DQ284	DecisionQuotient.ThermodynamicLift.energyLowerBound
DQ285	DecisionQuotient.ThermodynamicLift.energy_lower_additive
DQ286	DecisionQuotient.ThermodynamicLift.energy_lower_from_bits_lower
DQ287	DecisionQuotient.ThermodynamicLift.energy_lower_mandatory
DQ288	DecisionQuotient.ThermodynamicLift.eventual_thermo_lift
DQ289	DecisionQuotient.ThermodynamicLift.hardness_thermo_bundle_conditional
DQ290	DecisionQuotient.ThermodynamicLift.mandatory_conserved_bundle_conditional
DQ291	DecisionQuotient.ThermodynamicLift.mandatory_cost_bundle
DQ292	DecisionQuotient.TreeStructured
DQ293	DecisionQuotient.UniverseDynamics
DQ294	DecisionQuotient.ValueQueryState
DQ295	DecisionQuotient.actions_card
DQ296	DecisionQuotient.adversarialOpt
DQ297	DecisionQuotient.adversarialOpt_eq_false_of_ne
DQ298	DecisionQuotient.adversarialOpt_eq_true_of_eq
DQ299	DecisionQuotient.agreeOn
DQ300	DecisionQuotient.agreeOn_flipAt
DQ301	DecisionQuotient.agreeOn_iff_subset_agreementSet
DQ302	DecisionQuotient.agreeOn_xCoords_iff

ID	Full Lean handle (continued)
DQ303	DecisionQuotient.agreementSet
DQ304	DecisionQuotient.all_coords_relevant_of_not_tautology
DQ305	DecisionQuotient.anchor_sufficiency_sigma2p
DQ306	DecisionQuotient.anchor_sufficiency_sigma2p_nonempty
DQ307	DecisionQuotient.anchoredSlice
DQ308	DecisionQuotient.anchoredSliceEquivOutside
DQ309	DecisionQuotient.anchoredSlice_mul_fixed_eq_full
DQ310	DecisionQuotient.approxWithin
DQ311	DecisionQuotient.bestExpectedUtility
DQ312	DecisionQuotient.bool_minimalSufficient_unique
DQ313	DecisionQuotient.bool_sufficient_erase_irrelevant
DQ314	DecisionQuotient.bounded_actions_complexity
DQ315	DecisionQuotient.bounded_actions_polynomial_time
DQ316	DecisionQuotient.buildDQProblem
DQ317	DecisionQuotient.card_anchoredSlice
DQ318	DecisionQuotient.card_anchoredSlice_eq_pow_sub
DQ319	DecisionQuotient.card_anchoredSlice_eq_uniform
DQ320	DecisionQuotient.card_binary_state
DQ321	DecisionQuotient.card_function_space
DQ322	DecisionQuotient.card_outside_eq_sub
DQ323	DecisionQuotient.certificate_lower_bound_for_I
DQ324	DecisionQuotient.certificate_lower_bound_for_I_card
DQ325	DecisionQuotient.certificate_lower_bound_for_I_card_summary
DQ326	DecisionQuotient.certificate_lower_bound_for_I_empty
DQ327	DecisionQuotient.certificate_lower_bound_for_I_empty_summary
DQ328	DecisionQuotient.certificate_lower_bound_for_I_summary
DQ329	DecisionQuotient.certificate_lower_bound_poly
DQ330	DecisionQuotient.certificate_lower_bound_poly_ge
DQ331	DecisionQuotient.certificate_lower_bound_poly_ge_summary
DQ332	DecisionQuotient.certificate_lower_bound_poly_summary
DQ333	DecisionQuotient.compatibleStates
DQ334	DecisionQuotient.complexity_summary
DQ335	DecisionQuotient.constTrueFinite_empty_sufficient
DQ336	DecisionQuotient.constTrueProblem
DQ337	DecisionQuotient.constTrueProblemFinite
DQ338	DecisionQuotient.constTrueProblemFinite_opt
DQ339	DecisionQuotient.constTrueProblem_opt
DQ340	DecisionQuotient.constTrue_empty_sufficient
DQ341	DecisionQuotient.const_vs_scaled_opt_view_equal
DQ342	DecisionQuotient.const_vs_scaled_value_entry_diff_at_true
DQ343	DecisionQuotient.constant_opt_no_relevant
DQ344	DecisionQuotient.constant_opt_quotientSize_one
DQ345	DecisionQuotient.countSatisfyingAssignments
DQ346	DecisionQuotient.countedCheckPairs
DQ347	DecisionQuotient.countedCheckPairs_steps_bound
DQ348	DecisionQuotient.countedCompare
DQ349	DecisionQuotient.countedOptEqual
DQ350	DecisionQuotient.covers
DQ351	DecisionQuotient.covers_iff_agreeOn
DQ352	DecisionQuotient.cube_lt_pow
DQ353	DecisionQuotient.cubic_step_bound
DQ354	DecisionQuotient.cyclic_dependencies_coNP_hard
DQ355	DecisionQuotient.decisionQuotient_cases
DQ356	DecisionQuotient.decision_quotient_FPTAS
DQ357	DecisionQuotient.decision_quotient_sharp_P_hard

ID	Full Lean handle (continued)
DQ358	DecisionQuotient.decode_error_sum_two_labels
DQ359	DecisionQuotient.dichotomy_by_relevant_size
DQ360	DecisionQuotient.distinguish_requires_target
DQ361	DecisionQuotient.dqExact
DQ362	DecisionQuotient.dqProjection
DQ363	DecisionQuotient.dq_approximation_hard
DQ364	DecisionQuotient.emptySufficiency_query_indistinguishable_pair
DQ365	DecisionQuotient.emptySufficiency_query_indistinguishable_pair_finite
DQ366	DecisionQuotient.emptySufficiency_stateBatch_indistinguishable_pair
DQ367	DecisionQuotient.emptySufficiency_valueEntry_indistinguishable_pair
DQ368	DecisionQuotient.empty_minimal_sufficient_iff_constant
DQ369	DecisionQuotient.endpoints
DQ370	DecisionQuotient.endpoints_card_le
DQ371	DecisionQuotient.eth_implies_sat3_exponential
DQ372	DecisionQuotient.eth_lower_bound_informal
DQ373	DecisionQuotient.eth_lower_bound_sufficiency_check
DQ374	DecisionQuotient.exactDQ
DQ375	DecisionQuotient.exists_coord_not_mem
DQ376	DecisionQuotient.exists_distinct_patterns
DQ377	DecisionQuotient.exists_forall_iff_anchor_sufficient
DQ378	DecisionQuotient.exists_forall_iff_anchor_sufficient_padded
DQ379	DecisionQuotient.exists_not_mem_of_card_lt_univ
DQ380	DecisionQuotient.exists_state_not_in_endpoints
DQ381	DecisionQuotient.expectedUtility
DQ382	DecisionQuotient.exponential_query_complexity
DQ383	DecisionQuotient.feasibleActions
DQ384	DecisionQuotient.flipAt
DQ385	DecisionQuotient.flipAt_ne
DQ386	DecisionQuotient.fptRunningTime
DQ387	DecisionQuotient.fpt_kernel_bound
DQ388	DecisionQuotient.full_query_distinguishes_const_spike_finite
DQ389	DecisionQuotient.greedy_approximation_ratio
DQ390	DecisionQuotient.hard_when_all_relevant
DQ391	DecisionQuotient.indistinguishable_pair_forces_one_error
DQ392	DecisionQuotient.indistinguishable_pair_forces_one_error_per_seed
DQ393	DecisionQuotient.infeasible_not_optimal_of_gap
DQ394	DecisionQuotient.lawDecisionProblem
DQ395	DecisionQuotient.lawUtility
DQ396	DecisionQuotient.lawUtility_eq_of_allowed_iff
DQ397	DecisionQuotient.logicallyDeterministic
DQ398	DecisionQuotient.mem_compatibleStates_iff
DQ399	DecisionQuotient.mem_optFinset_iff
DQ400	DecisionQuotient.mem_optimalActions_iff_actionValue
DQ401	DecisionQuotient.mem_xCoords
DQ402	DecisionQuotient.min_sufficient_W2_hard
DQ403	DecisionQuotient.min_sufficient_inapproximability_informal
DQ404	DecisionQuotient.min_sufficient_set_coNP_hard
DQ405	DecisionQuotient.min_sufficient_set_inapprox_statement
DQ406	DecisionQuotient.mkState
DQ407	DecisionQuotient.mkState_castAdd
DQ408	DecisionQuotient.mkState_natAdd
DQ409	DecisionQuotient.monotone_opt_at_top
DQ410	DecisionQuotient.mss_equiv_relevant_card
DQ411	DecisionQuotient.no_satisfying_of_count_zero
DQ412	DecisionQuotient.not_tautology_iff_exists_opt_difference

ID	Full Lean handle (continued)
DQ413	DecisionQuotient.numPatterns
DQ414	DecisionQuotient.optComparisonCost
DQ415	DecisionQuotient.optFinset
DQ416	DecisionQuotient.optFinset_subset_projectedOptCover
DQ417	DecisionQuotient.opt_both
DQ418	DecisionQuotient.opt_eq_feasible_of_gap
DQ419	DecisionQuotient.opt_eq_of_allowed_iff
DQ420	DecisionQuotient.opt_falsifying
DQ421	DecisionQuotient.opt_falsifying_many
DQ422	DecisionQuotient.opt_no_only
DQ423	DecisionQuotient.opt_reference
DQ424	DecisionQuotient.opt_reference_many
DQ425	DecisionQuotient.opt_satisfying
DQ426	DecisionQuotient.opt_singleton_of_logicallyDeterministic
DQ427	DecisionQuotient.opt_tautology_many
DQ428	DecisionQuotient.opt_yes_only
DQ429	DecisionQuotient.optimalActions_eq_of_separable
DQ430	DecisionQuotient.optimalActions_sat
DQ431	DecisionQuotient.optimalActions_unsat
DQ432	DecisionQuotient.oracleView
DQ433	DecisionQuotient.oracleViewFinite
DQ434	DecisionQuotient.oracleViewFinite_eq_of_agree
DQ435	DecisionQuotient.oracleView_eq_of_agree
DQ436	DecisionQuotient.over_modeling_justified
DQ437	DecisionQuotient.pairCheckCost
DQ438	DecisionQuotient.parameterized_complexity_summary
DQ439	DecisionQuotient.patternClass
DQ440	DecisionQuotient.poly_compose_bound
DQ441	DecisionQuotient.poly_inner_bound
DQ442	DecisionQuotient.poly_reduction_implies_many_one_exists
DQ443	DecisionQuotient.poly_reduction_preserves_P
DQ444	DecisionQuotient.polytime_elicitation_exists_structured
DQ445	DecisionQuotient.proj_binary_state
DQ446	DecisionQuotient.projectToCoords
DQ447	DecisionQuotient.projectedOptCover
DQ448	DecisionQuotient.projectedOptCover_eq_opt_of_sufficient
DQ449	DecisionQuotient.qbfEval
DQ450	DecisionQuotient.qbfEval_mkState
DQ451	DecisionQuotient.qbfProblem
DQ452	DecisionQuotient.qbfUtility
DQ453	DecisionQuotient.queryComplexityLowerBound
DQ454	DecisionQuotient.queryDistinguishes
DQ455	DecisionQuotient.query_lower_bound_statement
DQ456	DecisionQuotient.quotientSize_bool_le_pow
DQ457	DecisionQuotient.quotientSize_le_card
DQ458	DecisionQuotient.quotientSize_le_pow_coords
DQ459	DecisionQuotient.quotientSize_one_all_sufficient
DQ460	DecisionQuotient.quotientSize_one_iff_constant
DQ461	DecisionQuotient.quotientSize_pos
DQ462	DecisionQuotient.recoverCount
DQ463	DecisionQuotient.recoverCount_correct
DQ464	DecisionQuotient.reductionProblem
DQ465	DecisionQuotient.reductionProblemMany
DQ466	DecisionQuotient.reductionUtility
DQ467	DecisionQuotient.reductionUtilityMany

ID	Full Lean handle (continued)
DQ468	DecisionQuotient.reduction_not_separable
DQ469	DecisionQuotient.replace_proj_other
DQ470	DecisionQuotient.replace_proj_self
DQ471	DecisionQuotient.sameProjection
DQ472	DecisionQuotient.sameProjection_trans
DQ473	DecisionQuotient.sat3ToCircuit
DQ474	DecisionQuotient.sat3_reduction_size_preserving
DQ475	DecisionQuotient.satSet
DQ476	DecisionQuotient.satSet_empty_of_count_zero
DQ477	DecisionQuotient.scaledTrueProblem
DQ478	DecisionQuotient.scaledTrueProblem_opt
DQ479	DecisionQuotient.separable_isSufficient
DQ480	DecisionQuotient.sharpSAT_encoded_in_DQ
DQ481	DecisionQuotient.sharpSAT_exactDQ
DQ482	DecisionQuotient.sharpSATtoDQ
DQ483	DecisionQuotient.sharpSATtoDQInstance
DQ484	DecisionQuotient.single_action_always_sufficient
DQ485	DecisionQuotient.someEmbedding
DQ486	DecisionQuotient.spikeFinite_empty_not_sufficient
DQ487	DecisionQuotient.spikeProblem
DQ488	DecisionQuotient.spikeProblemFinite
DQ489	DecisionQuotient.spikeProblemFinite_opt_at
DQ490	DecisionQuotient.spikeProblemFinite_opt_off
DQ491	DecisionQuotient.spikeProblem_opt_at
DQ492	DecisionQuotient.spikeProblem_opt_off
DQ493	DecisionQuotient.spike_empty_not_sufficient
DQ494	DecisionQuotient.stateBatchView
DQ495	DecisionQuotient.stateBatchView_eq_if_hidden_untouched
DQ496	DecisionQuotient.structured_isSufficient
DQ497	DecisionQuotient.succ_cube
DQ498	DecisionQuotient.sufficiencyToSetComparison
DQ499	DecisionQuotient.sufficiency_FPT_coords
DQ500	DecisionQuotient.sufficiency_W1_hard_unbounded_actions
DQ501	DecisionQuotient.sufficiency_check_coNP_hard
DQ502	DecisionQuotient.sufficiency_check_polynomial
DQ503	DecisionQuotient.sufficiency_iff_dq_ratio
DQ504	DecisionQuotient.sufficiency_iff_projectedOptCover_eq_opt
DQ505	DecisionQuotient.sufficiency_in_P
DQ506	DecisionQuotient.sufficiency_poly_bounded_actions
DQ507	DecisionQuotient.sufficiency_poly_separable
DQ508	DecisionQuotient.sufficiency_poly_tree_structured
DQ509	DecisionQuotient.sufficient_implies_tautology
DQ510	DecisionQuotient.sufficient_implies_tautology_many
DQ511	DecisionQuotient.sufficient_means_factorizable
DQ512	DecisionQuotient.sufficient_of_projectedOptCover_eq_opt
DQ513	DecisionQuotient.sufficient_preserves_decisions
DQ514	DecisionQuotient.targetProblem
DQ515	DecisionQuotient.targetProblem_opt_off_target
DQ516	DecisionQuotient.targetProblem_opt_on_target
DQ517	DecisionQuotient.targetProblems_agree_outside
DQ518	DecisionQuotient.tautology_iff_opt_constant
DQ519	DecisionQuotient.tautology_iff_sufficient
DQ520	DecisionQuotient.tautology_iff_sufficient_many
DQ521	DecisionQuotient.tautology_implies_sufficient
DQ522	DecisionQuotient.tautology_implies_sufficient_many

ID	Full Lean handle (continued)
DQ523	DecisionQuotient.totalCheckCost
DQ524	DecisionQuotient.totalCheckCost_le_pow
DQ525	DecisionQuotient.touchedStates
DQ526	DecisionQuotient.touchedStates_card_le_query_card
DQ527	DecisionQuotient.tractability_conditions_tight
DQ528	DecisionQuotient.tractable_small_state_space
DQ529	DecisionQuotient.tractable_when_few_relevant
DQ530	DecisionQuotient.two_actions_coNP_hard
DQ531	DecisionQuotient.two_le_one_false
DQ532	DecisionQuotient.two_le_zero_false
DQ533	DecisionQuotient.two_state_sufficiency
DQ534	DecisionQuotient.unique_matching_pattern
DQ535	DecisionQuotient.universe_sets_objective_schema
DQ536	DecisionQuotient.valueEntryView
DQ537	DecisionQuotient.valueEntryView_eq_if_hidden_untouched
DQ538	DecisionQuotient.valueEntryView_eq_of_stateBatchView_eq_on_touched
DQ539	DecisionQuotient.valueOfInformation
DQ540	DecisionQuotient.valueOfInformation_const
DQ541	DecisionQuotient.vectorB_blocked
DQ542	DecisionQuotient.voi_computation_sharp_P_hard
DQ543	DecisionQuotient.voi_fptas_smooth_utilities
DQ544	DecisionQuotient.weightedQueryCost
DQ545	DecisionQuotient.weightedQueryCost_ge_min_mul_card
DQ546	DecisionQuotient.weightedQueryCost_ge_min_mul_of_card_lb
DQ547	DecisionQuotient.weighted_seed_error_identity
DQ548	DecisionQuotient.weighted_seed_half_floor
DQ549	DecisionQuotient.xCoords
DQ550	DecisionQuotient.xPart
DQ551	DecisionQuotient.xPart_mkState
DQ552	DecisionQuotient.yAllFalse
DQ553	DecisionQuotient.yAllFalse_mkState
DQ554	DecisionQuotient.yPart
DQ555	DecisionQuotient.yPart_mkState
DQ556	DecisionQuotient.Physics.Instantiation.Circuit
DQ557	DecisionQuotient.Physics.Instantiation.Dynamics
DQ558	DecisionQuotient.Physics.Instantiation.Geometry
DQ559	DecisionQuotient.Physics.Instantiation.MoleculeCircuit
DQ560	DecisionQuotient.Physics.Instantiation.MoleculeDynamics
DQ561	DecisionQuotient.Physics.Instantiation.MoleculeGeometry
DQ562	DecisionQuotient.Physics.Instantiation.geometry_plus_dynamics_is_circuit
DQ563	DecisionQuotient.Physics.Instantiation.law_objective_schema
DQ564	DecisionQuotient.Physics.Instantiation.law_opt_eq_feasible_of_gap
DQ565	DecisionQuotient.Physics.Instantiation.law_opt_singleton_of_deterministic
F1	Formula.eval
F2	Formula.isTautology
F3	Formula.majorityTrue
GC1	GoalClass.classMonotoneOn
GC2	GoalClass.isNonNegativelyTautologicalCoord
GC3	GoalClass.isTautologicalCoord
GC4	GoalClass.tautologicalSet
HD1	DecisionQuotient.HardnessDistribution.centralization_dominance_bundle
HD2	DecisionQuotient.HardnessDistribution.centralization_step_saves_n_minus_one
HD3	DecisionQuotient.HardnessDistribution.centralized_higher_leverage
HD4	DecisionQuotient.HardnessDistribution.complete_model_dominates_after_threshold
HD5	DecisionQuotient.HardnessDistribution.gap_conservation_card

ID	Full Lean handle (continued)
HD6	DecisionQuotient.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant
HD7	DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness
HD8	DecisionQuotient.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
HD9	DecisionQuotient.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
HD10	DecisionQuotient.HardnessDistribution.generalized_right_eventually_dominates_wrong
HD11	DecisionQuotient.HardnessDistribution.hardnessEfficiency_eq_central_share
HD12	DecisionQuotient.HardnessDistribution.isRightHardness
HD13	DecisionQuotient.HardnessDistribution.isWrongHardness
HD14	DecisionQuotient.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
HD15	DecisionQuotient.HardnessDistribution.native_dominates_manual
HD16	DecisionQuotient.HardnessDistribution.no_positive_slope_linear_represents_saturating
HD17	DecisionQuotient.HardnessDistribution.right_dominates_wrong
HD18	DecisionQuotient.HardnessDistribution.saturatingSiteCost_eventually_constant
HD19	DecisionQuotient.HardnessDistribution.simplicityTax_grows
HD20	DecisionQuotient.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
HD21	DecisionQuotient.HardnessDistribution.totalDOF_ge_intrinsic
IC1	DecisionQuotient.IntegrityCompetence.CertaintyInflation
IC2	DecisionQuotient.IntegrityCompetence.CompletionFractionDefined
IC3	DecisionQuotient.IntegrityCompetence.EvidenceForReport
IC4	DecisionQuotient.IntegrityCompetence.ExactCertaintyInflation
IC5	DecisionQuotient.IntegrityCompetence.Percent
IC6	DecisionQuotient.IntegrityCompetence.RLFFWeights
IC7	DecisionQuotient.IntegrityCompetence.ReportSignal
IC8	DecisionQuotient.IntegrityCompetence.ReportBitModel
IC9	DecisionQuotient.IntegrityCompetence.SignalConsistent
IC10	DecisionQuotient.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
IC11	DecisionQuotient.IntegrityCompetence.admissible_matrix_counts
IC12	DecisionQuotient.IntegrityCompetence.abstain_signal_exists_with_guess_self
IC13	DecisionQuotient.IntegrityCompetence.certaintyInflation_iff_not_admissible
IC14	DecisionQuotient.IntegrityCompetence.certificationOverheadBits
IC15	DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_evidence
IC16	DecisionQuotient.IntegrityCompetence.certificationOverheadBits_of_no_evidence
IC17	DecisionQuotient.IntegrityCompetence.certifiedTotalBits
IC18	DecisionQuotient.IntegrityCompetence.certifiedTotalBits_ge_raw
IPT1	IsPolynomialTime.const
IPT2	IsPolynomialTime.of_steps_le
IW1	InsufficiencyWitness.certifies_not_sufficient
IW2	InsufficiencyWitness.not_abstract_sufficient
L1	ExactPhysicalClaimWellTyped
L2	ExcusedBy
L3	ExplicitHardnessAssumptions
L4	Literal.eval
L5	OneStepSequentialObjective
L6	RegimeSimulation
L7	StandardComplexityAssumptions
L8	StochasticCriterionObjective
L9	StructureDetectable
L10	TransitionCoupledObjective
L11	TwoStepObjective
L12	adq
L13	adq_ordering
L14	agentBridgeClass

ID	Full Lean handle (continued)
L15	agent_transfer_licensed_iff_snapshot
L16	anchor_sigma2p_complete_conditional
L17	anchor_sigma2p_reduction_core
L18	boundaryCharacterized
L19	boundaryCharacterized_iff_exists_sufficient_subset
L20	bounded_actions_detectable
L21	bounded_actions_reusable_heuristic
L22	bridgeFailureWitness
L23	boolHypercube_node_count
L24	bridgeTransferLicensed
L25	bridge_boundary_represented_family
L26	bridge_failure_witness_non_one_step
L27	bridge_transfer_iff_one_step_class
L28	certified_total_bits_split_core
L29	constantBoolDP
L30	constantBoolDP_empty_sufficient
L31	constantBoolDP_opt
L32	cost_asymmetry_eth_conditional
L33	declaredRegimeFamily
L34	declaredRegimeFamily_complete
L35	declared_physics_no_universal_exact_certifier_core
L36	dichotomy_conditional
L37	epsilon_admissible_iff_raw_lt_certified_total_core
L38	exact_admissible_iff_raw_lt_certified_total_core
L39	exact_certainty_inflation_under_hardness_core
L40	exact_raw_eq_certified_iff_certainty_inflation_core
L41	exact_raw_only_of_no_exact_admissible_core
L42	explicit_assumptions_required_of_not_excused_core
L43	explicit_state_upper_core
L44	firstCoordDP
L45	firstCoordDP_empty_not_sufficient
L46	firstCoordDP_opt
L47	hard_family_all_coords_core
L48	horizonTwoWitness
L49	horizonTwoWitness_immediate_empty_sufficient
L50	horizonTwoWitness_not_empty_sufficient_two_step
L51	horizon_gt_one_bridge_can_fail_on_sufficiency
L52	i0
L53	i0_bridge
L54	information_barrier_opt_oracle_core
L55	information_barrier_state_batch_core
L56	information_barrier_value_entry_core
L57	integrity_resource_bound_for_sufficiency
L58	integrity_universal_applicability_core
L59	minsuff_collapse_core
L60	minsuff_collapse_to_conp_conditional
L61	minsuff_conp_complete_conditional
L62	no_auto_minimize_of_p_neq_conp
L63	no_exact_claim_admissible_under_hardness_core
L64	no_exact_claim_under_declared_assumptions_unless_excused_core
L65	no_exact_identifier_implies_not_boundary_characterized
L66	no_uncertified_exact_claim_core
L67	node_count_does_not_determine_edge_geometry
L68	one_step_bridge
L69	oracle_lattice_transfer_as_regime_simulation

ID	Full Lean handle (continued)
L70	physical_crossover_above_cap_core
L71	physical_crossover_core
L72	physical_crossover_hardness_core
L73	physical_crossover_policy_core
L74	preferTrueSelector
L75	process_bridge_failure_witness
L76	query_obstruction_boolean_corollary
L77	query_obstruction_finite_state_core
L78	regimeCoreClaim
L79	regime_core_claim_proved
L80	regime_simulation_transfers_hardness
L81	reusable_heuristic_of_detectable
L82	s0_bridge
L83	s1_bridge
L84	sFalse
L85	sTrue
L86	selectorGapProblem
L87	selectorGap_not_set_sufficient_empty
L88	selectorGap_selectorSufficient_empty
L89	selectorGap_true_mem_opt
L90	selectorSufficient_not_implies_setSufficient
L91	separable_detectable
L92	separable_reusable_heuristic
L93	snapshot_vs_process_typed_boundary
L94	standard_assumption_ledger_unpack
L95	stochasticWitness
L96	stochastic_objective_bridge_can_fail_on_sufficiency
L97	structureDetectable_of_decidable
L98	subproblem_hardness_lifts_to_full
L99	subproblem_transfer_as_regime_simulation
L100	sufficiency_conp_complete_conditional
L101	sufficiency_conp_reduction_core
L102	thermo_conservation_additive_core
L103	thermo_energy_carbon_lift_core
L104	thermo_eventual_lift_core
L105	thermo_hardness_bundle_core
L106	thermo_mandatory_cost_core
L107	tractable_bounded_core
L108	tractable_separable_core
L109	tractable_subcases_conditional
L110	tractable_tree_core
L111	transitionWitness
L112	transition_coupled_bridge_can_fail_on_sufficiency
L113	tree_reusable_heuristic
L114	tree_structure_detectable
L115	typed_claim_admissibility_core
L116	typed_static_class_completeness
L117	universal_solver_framing_core
MC1	MatrixCell.verdict
MC2	MatrixCell.verdict_substrate_independent
OSSO1	OneStepSequentialObjective.Opt
OSSO2	OneStepSequentialObjective.isSufficient
OSSO3	OneStepSequentialObjective.toDecisionProblem
P4SS1	Paper4bStochasticSequential.Assignment
P4SS10	Paper4bStochasticSequential.ClaimClosure.claim_trajectory_substrate_dependent

ID	Full Lean handle (continued)
P4SS100	Paper4bStochasticSequential.bounded_horizon_tractable_rigorous
P4SS101	Paper4bStochasticSequential.bounded_support_tractable_rigorous
P4SS102	Paper4bStochasticSequential.card_stochAction
P4SS103	Paper4bStochasticSequential.card_stochState
P4SS104	Paper4bStochasticSequential.coinFlips
P4SS105	Paper4bStochasticSequential.coinFlips_product
P4SS106	Paper4bStochasticSequential.composeReduction4b
P4SS107	Paper4bStochasticSequential.costlySignal
P4SS108	Paper4bStochasticSequential.countedBoundedHorizonSufficiency
P4SS109	Paper4bStochasticSequential.countedBoundedHorizonSufficiency_steps
P4SS11	Paper4bStochasticSequential.ClaimClosure.claim_verdict_substrate_independent
P4SS110	Paper4bStochasticSequential.countedBoundedSupportCheck
P4SS111	Paper4bStochasticSequential.countedBoundedSupportCheck_steps
P4SS112	Paper4bStochasticSequential.countedExpectedUtility
P4SS113	Paper4bStochasticSequential.countedExpectedUtility_steps
P4SS114	Paper4bStochasticSequential.countedIsOptimal
P4SS115	Paper4bStochasticSequential.countedIsOptimal_steps
P4SS116	Paper4bStochasticSequential.countedIsRelevantCoord
P4SS117	Paper4bStochasticSequential.countedIsRelevantCoord_steps
P4SS118	Paper4bStochasticSequential.countedProductSufficiencyCheck
P4SS119	Paper4bStochasticSequential.countedProductSufficiencyCheck_steps
P4SS12	Paper4bStochasticSequential.ClaimClosure.timeComplexity'
P4SS120	Paper4bStochasticSequential.countedStochOptEqual
P4SS121	Paper4bStochasticSequential.countedStochSufficiency
P4SS122	Paper4bStochasticSequential.countedStochSufficiency_poly
P4SS123	Paper4bStochasticSequential.countedStochasticOpt
P4SS124	Paper4bStochasticSequential.countedStochasticOpt_steps
P4SS125	Paper4bStochasticSequential.critical_value_half
P4SS126	Paper4bStochasticSequential.decisionEntropy
P4SS127	Paper4bStochasticSequential.detectSufficiency
P4SS128	Paper4bStochasticSequential.dichotomyThreshold
P4SS129	Paper4bStochasticSequential.diminishing_returns
P4SS13	Paper4bStochasticSequential.ClaimClosure.trajectory
P4SS130	Paper4bStochasticSequential.distributionSupport
P4SS131	Paper4bStochasticSequential.energyTimeTradeoff
P4SS132	Paper4bStochasticSequential.energyUsed
P4SS133	Paper4bStochasticSequential.entropy
P4SS134	Paper4bStochasticSequential.equilibriumPrice
P4SS135	Paper4bStochasticSequential.evidenceMonotoneRetraction
P4SS136	Paper4bStochasticSequential.executeCountedQuery
P4SS137	Paper4bStochasticSequential.executeStrategy
P4SS138	Paper4bStochasticSequential.expectedQueryComplexity
P4SS139	Paper4bStochasticSequential.expectedUtilityAfterObservation
P4SS14	Paper4bStochasticSequential.ClaimSequence
P4SS140	Paper4bStochasticSequential.expectedUtility_via_marginals
P4SS141	Paper4bStochasticSequential.expectedValue
P4SS142	Paper4bStochasticSequential.extractPolicy
P4SS143	Paper4bStochasticSequential.extractableWork
P4SS144	Paper4bStochasticSequential.full_compression_when_sufficient
P4SS145	Paper4bStochasticSequential.fully_observable_tractable_rigorous
P4SS146	Paper4bStochasticSequential.gamblerSequential
P4SS147	Paper4bStochasticSequential.hardFamily
P4SS148	Paper4bStochasticSequential.hard_family_complexity
P4SS149	Paper4bStochasticSequential.heatDissipated
P4SS15	Paper4bStochasticSequential.ClaimWithEvidence

ID	Full Lean handle (continued)
P4SS150	Paper4bStochasticSequential.heuristic_transfer
P4SS151	Paper4bStochasticSequential.horizon_gt_one_can_fail
P4SS152	Paper4bStochasticSequential.info_sufficient_implies_stochastic
P4SS153	Paper4bStochasticSequential.informationSufficient
P4SS154	Paper4bStochasticSequential.integrity_competence_verdict_independent
P4SS155	Paper4bStochasticSequential.inventoryMDP
P4SS156	Paper4bStochasticSequential.inventoryMDP_fully_observable
P4SS157	Paper4bStochasticSequential.isProbabilityDistribution
P4SS158	Paper4bStochasticSequential.isProductDistribution
P4SS159	Paper4bStochasticSequential.isRelevantCoord
P4SS16	Paper4bStochasticSequential.ComplexityClass
P4SS160	Paper4bStochasticSequential.kappa_sufficient_statistic
P4SS161	Paper4bStochasticSequential.landauerConstant
P4SS162	Paper4bStochasticSequential.landauer_limit
P4SS163	Paper4bStochasticSequential.majsat_implies_sufficient
P4SS164	Paper4bStochasticSequential.marginalValue
P4SS165	Paper4bStochasticSequential.maxExpectedUtility
P4SS166	Paper4bStochasticSequential.mdpValueIteration
P4SS167	Paper4bStochasticSequential.mdpValueIterationStep
P4SS168	Paper4bStochasticSequential.mdpValueIterationStep_steps
P4SS169	Paper4bStochasticSequential.mdpValueIteration_steps
P4SS17	Paper4bStochasticSequential.CountedQueryAlgorithm
P4SS170	Paper4bStochasticSequential.more_energy_faster
P4SS171	Paper4bStochasticSequential.mutualInfoCoordAction
P4SS172	Paper4bStochasticSequential.noRetractionWithoutEvidence
P4SS173	Paper4bStochasticSequential.no_arbitrage
P4SS174	Paper4bStochasticSequential.optimalAlgo'
P4SS175	Paper4bStochasticSequential.perturbationDistribution
P4SS176	Paper4bStochasticSequential.perturbedStochProblem
P4SS177	Paper4bStochasticSequential.phaseTransitionParam
P4SS178	Paper4bStochasticSequential.policyFromHistory
P4SS179	Paper4bStochasticSequential.posterior
P4SS18	Paper4bStochasticSequential.Distribution
P4SS180	Paper4bStochasticSequential.posterior_uniform_prior
P4SS181	Paper4bStochasticSequential.pp_decide_correct
P4SS182	Paper4bStochasticSequential.probSufficient
P4SS183	Paper4bStochasticSequential.product_distribution_tractable_rigorous
P4SS184	Paper4bStochasticSequential.query_computation_tradeoff
P4SS185	Paper4bStochasticSequential.query_lower_bound_insufficient
P4SS186	Paper4bStochasticSequential.query_lower_bound_sufficient
P4SS187	Paper4bStochasticSequential.randomStochProblem
P4SS188	Paper4bStochasticSequential.reduceMAJSAT
P4SS189	Paper4bStochasticSequential.reduceMAJSAT_correct
P4SS19	Paper4bStochasticSequential.Distribution.delta
P4SS190	Paper4bStochasticSequential.reduceMAJSAT_hard
P4SS191	Paper4bStochasticSequential.reduceMAJSAT_poly
P4SS192	Paper4bStochasticSequential.reduceMAJSAT_polytime
P4SS193	Paper4bStochasticSequential.reduceTQBF
P4SS194	Paper4bStochasticSequential.reduceTQBF_correct
P4SS195	Paper4bStochasticSequential.reduceTQBF_hard
P4SS196	Paper4bStochasticSequential.reduceTQBF_poly
P4SS197	Paper4bStochasticSequential.reduction_hardness_transfer
P4SS198	Paper4bStochasticSequential.refinement_strengthens
P4SS199	Paper4bStochasticSequential.regime_hierarchy
P4SS2	Paper4bStochasticSequential.BigO

ID	Full Lean handle (continued)
P4SS20	Paper4bStochasticSequential.Distribution.probability
P4SS200	Paper4bStochasticSequential.relevantCoordSize
P4SS201	Paper4bStochasticSequential.relevant_coord_concentration
P4SS202	Paper4bStochasticSequential.retraction_with_evidence_preserves
P4SS203	Paper4bStochasticSequential.retraction_without_evidence_violates
P4SS204	Paper4bStochasticSequential.reversibleSufficiencyAlgo
P4SS205	Paper4bStochasticSequential.reversible_lower_cost
P4SS206	Paper4bStochasticSequential.seqProblem
P4SS207	Paper4bStochasticSequential.sequentialDecisionEquiv
P4SS208	Paper4bStochasticSequential.sequentialQueryComplexity
P4SS209	Paper4bStochasticSequential.sequentialValue
P4SS21	Paper4bStochasticSequential.Distribution.uniform
P4SS210	Paper4bStochasticSequential.sequential_complexity
P4SS211	Paper4bStochasticSequential.sequential_not_stochastic
P4SS212	Paper4bStochasticSequential.sequential_query_bounded_horizon
P4SS213	Paper4bStochasticSequential.sequential_sufficiency_pspace_complete
P4SS214	Paper4bStochasticSequential.sequential_sufficiency_pspace_hard
P4SS215	Paper4bStochasticSequential.sequential_sufficient_in_PSPACE
P4SS216	Paper4bStochasticSequential.signal_cost_lower_when_sufficient
P4SS217	Paper4bStochasticSequential.smoothedComplexity
P4SS218	Paper4bStochasticSequential.smoothed_easier_than_worst
P4SS219	Paper4bStochasticSequential.someAction
P4SS22	Paper4bStochasticSequential.Distribution.uniformList
P4SS220	Paper4bStochasticSequential.static_complexity
P4SS221	Paper4bStochasticSequential.static_subset_stochastic
P4SS222	Paper4bStochasticSequential.static_to_sequential_transfer
P4SS223	Paper4bStochasticSequential.static_to_stochastic_can_fail
P4SS224	Paper4bStochasticSequential.static_to_stochastic_transfer
P4SS225	Paper4bStochasticSequential.stochDistribution
P4SS226	Paper4bStochasticSequential.stochProblem
P4SS227	Paper4bStochasticSequential.stochUtility
P4SS228	Paper4bStochasticSequential.stoch_opt_falsifying
P4SS229	Paper4bStochasticSequential.stoch_opt_reference
P4SS23	Paper4bStochasticSequential.Evidence
P4SS230	Paper4bStochasticSequential.stoch_opt_satisfying
P4SS231	Paper4bStochasticSequential.stochaEnumAlgo
P4SS232	Paper4bStochasticSequential.stochaEnumAlgo_optimal
P4SS233	Paper4bStochasticSequential.stochaPPDecide
P4SS234	Paper4bStochasticSequential.stochasticCompressionRatio
P4SS235	Paper4bStochasticSequential.stochasticDecisionEquiv
P4SS236	Paper4bStochasticSequential.stochasticExpectedUtility
P4SS237	Paper4bStochasticSequential.stochastic_complexity
P4SS238	Paper4bStochasticSequential.stochastic_few_relevant_tractable
P4SS239	Paper4bStochasticSequential.stochastic_many_relevant_hard
P4SS24	Paper4bStochasticSequential.Identifiable
P4SS240	Paper4bStochasticSequential.stochastic_not_static
P4SS241	Paper4bStochasticSequential.stochastic_query_product
P4SS242	Paper4bStochasticSequential.stochastic_quotient_factors
P4SS243	Paper4bStochasticSequential.stochastic_sufficiency_pp_complete
P4SS244	Paper4bStochasticSequential.stochastic_sufficiency_pp_hard
P4SS245	Paper4bStochasticSequential.stochastic_sufficient_in_PP
P4SS246	Paper4bStochasticSequential.stochastic_sufficient_not_info
P4SS247	Paper4bStochasticSequential.stochastic_to_sequential_can_fail
P4SS248	Paper4bStochasticSequential.stochastic_to_sequential_transfer
P4SS249	Paper4bStochasticSequential.substrateCost

ID	Full Lean handle (continued)
P4SS25	Paper4bStochasticSequential.InComplexityClass
P4SS250	Paper4bStochasticSequential.substrate_independence_verdict
P4SS251	Paper4bStochasticSequential.sufficient_enables_work_extraction
P4SS252	Paper4bStochasticSequential.sufficient_implies_majsat
P4SS253	Paper4bStochasticSequential.temporalIntegrity
P4SS254	Paper4bStochasticSequential.temporal_integrity_hardness
P4SS255	Paper4bStochasticSequential.thermodynamicCost
P4SS256	Paper4bStochasticSequential.timeComplexity
P4SS257	Paper4bStochasticSequential.timeComplexity''
P4SS258	Paper4bStochasticSequential.toStatic
P4SS259	Paper4bStochasticSequential.toStochastic
P4SS26	Paper4bStochasticSequential.IsPolyTime
P4SS260	Paper4bStochasticSequential.trajectory
P4SS261	Paper4bStochasticSequential.trajectory_substrate_dependent
P4SS262	Paper4bStochasticSequential.umbrellaStatic
P4SS263	Paper4bStochasticSequential.umbrellaStochastic
P4SS264	Paper4bStochasticSequential.umbrella_expected_carry
P4SS265	Paper4bStochasticSequential.uniformDistribution
P4SS266	Paper4bStochasticSequential.uniformStochDistribution
P4SS267	Paper4bStochasticSequential.uniformStochDistribution_valid
P4SS268	Paper4bStochasticSequential.uniform_is_valid
P4SS269	Paper4bStochasticSequential.valueIterStep
P4SS27	Paper4bStochasticSequential.MAJSAT_pp_complete
P4SS270	Paper4bStochasticSequential.valueIteration
P4SS271	Paper4bStochasticSequential.valueIteration_poly
P4SS272	Paper4bStochasticSequential.valueOfInformation
P4SS273	Paper4bStochasticSequential.voi_positive_when_insufficient
P4SS274	Paper4bStochasticSequential.voi_zero_when_sufficient
P4SS28	Paper4bStochasticSequential.MAJSAT_pred
P4SS29	Paper4bStochasticSequential.MatrixCell
P4SS3	Paper4bStochasticSequential.ClaimClosure.PP_bound
P4SS30	Paper4bStochasticSequential.PP_class
P4SS31	Paper4bStochasticSequential.PSPACE_CLASS
P4SS32	Paper4bStochasticSequential.PSPACE_class
P4SS33	Paper4bStochasticSequential.PTIME
P4SS34	Paper4bStochasticSequential.Policy
P4SS35	Paper4bStochasticSequential.PolyReduction4b
P4SS36	Paper4bStochasticSequential.PolynomialSpaceComputable
P4SS37	Paper4bStochasticSequential.PolynomialTimeComputable
P4SS38	Paper4bStochasticSequential.ProblemSequence
P4SS39	Paper4bStochasticSequential.ProductDistributionStructure
P4SS4	Paper4bStochasticSequential.ClaimClosure.claim_hierarchy
P4SS40	Paper4bStochasticSequential.QueryOracle
P4SS41	Paper4bStochasticSequential.QueryStrategy
P4SS42	Paper4bStochasticSequential.SequentialDecisionProblem
P4SS43	Paper4bStochasticSequential.SequentialQuotientType
P4SS44	Paper4bStochasticSequential.SequentialRegime
P4SS45	Paper4bStochasticSequential.SequentialSufficient
P4SS46	Paper4bStochasticSequential.StaticRegime
P4SS47	Paper4bStochasticSequential.StochInsufficiencyWitness
P4SS48	Paper4bStochasticSequential.StochasticDecisionProblem
P4SS49	Paper4bStochasticSequential.StochasticQuotientType
P4SS5	Paper4bStochasticSequential.ClaimClosure.claim_many_relevant_hard
P4SS50	Paper4bStochasticSequential.StochasticRegime
P4SS51	Paper4bStochasticSequential.StochasticSufficient

ID	Full Lean handle (continued)
P4SS52	Paper4bStochasticSequential.StructureClass
P4SS53	Paper4bStochasticSequential.StructurePrior
P4SS54	Paper4bStochasticSequential.TQBF
P4SS55	Paper4bStochasticSequential.TQBF_pspace_complete
P4SS56	Paper4bStochasticSequential.Tractability.PP_Complexity
P4SS57	Paper4bStochasticSequential.Tractability.PSPACE_Complexity
P4SS58	Paper4bStochasticSequential.Tractability.boundedHorizonAlgo
P4SS59	Paper4bStochasticSequential.Tractability.boundedSupportAlgo
P4SS6	Paper4bStochasticSequential.ClaimClosure.claim_phase_transition
P4SS60	Paper4bStochasticSequential.Tractability.bounded_horizon_polynomial
P4SS61	Paper4bStochasticSequential.Tractability.bounded_support_polynomial
P4SS62	Paper4bStochasticSequential.Tractability.bounded_support_tight
P4SS63	Paper4bStochasticSequential.Tractability.coordinateSufficient
P4SS64	Paper4bStochasticSequential.Tractability.f
P4SS65	Paper4bStochasticSequential.Tractability.fptTimeComplexity
P4SS66	Paper4bStochasticSequential.Tractability.fpt_horizon
P4SS67	Paper4bStochasticSequential.Tractability.fully_observable_polynomial
P4SS68	Paper4bStochasticSequential.Tractability.fully_observable_reduces_to_static
P4SS69	Paper4bStochasticSequential.Tractability.hasBoundedHorizon
P4SS7	Paper4bStochasticSequential.ClaimClosure.claim_static_sequential_horizon_one
P4SS70	Paper4bStochasticSequential.Tractability.hasBoundedSupport
P4SS71	Paper4bStochasticSequential.Tractability.isFullyObservable
P4SS72	Paper4bStochasticSequential.Tractability.isProductDistribution'
P4SS73	Paper4bStochasticSequential.Tractability.mdpAlgo
P4SS74	Paper4bStochasticSequential.Tractability.optimalAlgo
P4SS75	Paper4bStochasticSequential.Tractability.partially_observable_hard
P4SS76	Paper4bStochasticSequential.Tractability.productAlgo
P4SS77	Paper4bStochasticSequential.Tractability.product_decomposition
P4SS78	Paper4bStochasticSequential.Tractability.product_distribution_polynomial
P4SS79	Paper4bStochasticSequential.Tractability.product_distribution_tight
P4SS8	Paper4bStochasticSequential.ClaimClosure.claim_static_stochastic_product
P4SS80	Paper4bStochasticSequential.Tractability.remove_bounded_horizon_hard
P4SS81	Paper4bStochasticSequential.Tractability.remove_bounded_support_hard
P4SS82	Paper4bStochasticSequential.Tractability.remove_full_observability_hard
P4SS83	Paper4bStochasticSequential.Tractability.remove_product_distribution_hard
P4SS84	Paper4bStochasticSequential.Tractability.solvesSufficiency
P4SS85	Paper4bStochasticSequential.Tractability.timeComplexity
P4SS86	Paper4bStochasticSequential.Tractability.toStatic
P4SS87	Paper4bStochasticSequential.Tractability.tractability_conditions_minimal_sequential
P4SS88	Paper4bStochasticSequential.Tractability.tractability_conditions_minimal_stochastic
P4SS89	Paper4bStochasticSequential.Tractability.unbounded_horizon_hard
P4SS9	Paper4bStochasticSequential.ClaimClosure.claim_stochastic_sequential_memoryless
P4SS90	Paper4bStochasticSequential.above_critical_hard
P4SS91	Paper4bStochasticSequential.abstentionSet
P4SS92	Paper4bStochasticSequential.abstention_decreases
P4SS93	Paper4bStochasticSequential.achievable
P4SS94	Paper4bStochasticSequential.argmax
P4SS95	Paper4bStochasticSequential.averageCaseComplexity
P4SS96	Paper4bStochasticSequential.average_case_hard
P4SS97	Paper4bStochasticSequential.bayesian_update
P4SS98	Paper4bStochasticSequential.below_critical_polynomial
P4SS99	Paper4bStochasticSequential.boundedSupport
PBC1	DecisionQuotient.PhysicalBudgetCrossover.CrossoverAt
PBC2	DecisionQuotient.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
PC1	PhysicalComplexity.ComputationalRequirement

ID	Full Lean handle (continued)
PC2	PhysicalComplexity.InstanceSize
PC3	PhysicalComplexity.Landauer_bound
PC4	PhysicalComplexity.PhysicalComputer
PC5	PhysicalComplexity.bit_energy_cost
PC6	PhysicalComplexity.coNP_physically_impossible
PC7	PhysicalComplexity.coNP_requirement
PC8	PhysicalComplexity.k_Boltzmann
PC9	PhysicalComplexity.sufficiency_physically_impossible
PC10	PhysicalComplexity.AccessRegime.AccessRegime
PC11	PhysicalComplexity.AccessRegime.PhysicalDevice
PC12	PhysicalComplexity.AccessRegime.RegimeEval
PC13	PhysicalComplexity.AccessRegime.RegimeProof
PC14	PhysicalComplexity.AccessRegime.RegimeSample
PC15	PhysicalComplexity.AccessRegime.RegimeWithCertificate
PC16	PhysicalComplexity.AccessRegime.certificate_amortizes_hardness
PC17	PhysicalComplexity.AccessRegime.certificate_upgrades_regime
PC18	PhysicalComplexity.AccessRegime.physical_succinct_certification_hard
PC19	PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate
PC20	PhysicalComplexity.AccessRegime.AccessChannelLaw
PC21	PhysicalComplexity.AccessRegime.AuditableWithCertificate
PC22	PhysicalComplexity.AccessRegime.FiveWayMeet
PC23	PhysicalComplexity.AccessRegime.HardUnderEval
PC24	PhysicalComplexity.AccessRegime.RegimeEvalOn
PC25	PhysicalComplexity.AccessRegime.RegimeProofOn
PC26	PhysicalComplexity.AccessRegime.RegimeSampleOn
PC27	PhysicalComplexity.AccessRegime.RegimeWithCertificateOn
PC28	PhysicalComplexity.AccessRegime.certificate_upgrades_regime_on
PC29	PhysicalComplexity.AccessRegime.regime_upgrade_with_certificate_on
PC30	PhysicalComplexity.coNP_not_in_P_physically
PC31	PhysicalComplexity.pow2_eventually_exceeds
PD2	PhysicalDevice.is_succinct
PD3	PhysicalDevice.derived_explicit_size
PD4	PhysicalDevice.is_succinct_of_gap
PDS1	ProductDistributionStructure.toDistribution
PR1	PolyReduction.comp_exists
PR2	PolyReduction.id
QA1	QueryAlgorithm.run
QBF1	QBF.eval
QF1	QFormula.eval
QF2	QFormula.eval_map
QF3	QFormula.map
QT1	QueryTranscript.size
S1	Signal.const
S2	Signal.priorOn
SAT3I1	SAT3Instance.inputSize
SCO1	StochasticCriterionObjective.OptChance
SCO2	StochasticCriterionObjective.toExpectedDecisionProblem
SDP1	StochasticDecisionProblem.stochasticOpt
SR1	StaticRegime.toSequential
SR2	StaticRegime.toStochastic
TCO1	TransitionCoupledObjective.Opt
TCO2	TransitionCoupledObjective.score
TCO3	TransitionCoupledObjective.toImmediateDecisionProblem
TSO1	TwoStepObjective.Opt
TSO2	TwoStepObjective.score

ID	Full Lean handle (continued)
TSO3	TwoStepObjective.toImmediateDecisionProblem

11.2 What Is Machine-Checked

The Lean formalization establishes:

1. **STOCHASTIC-SUFFICIENCY PP-completeness**: Reduction from MAJSAT showing stochastic sufficiency checking is PP-complete.
2. **SEQUENTIAL-SUFFICIENCY PSPACE-completeness**: Reduction from TQBF showing sequential sufficiency checking is PSPACE-complete.
3. **Substrate independence theorems**: Formal proofs that the matrix verdict is κ-independent while trajectories are κ-dependent.
4. **Bridge transfer conditions**: Formalized conditions under which sufficiency transfers between regimes.

11.3 Assumption Ledger (Auto)

Source files.

- `Paper4bStochasticSequential/ClaimClosure.lean`

Assumption bundles and fields.

- (No assumption bundles parsed.)

Conditional closure handles.

- (No conditional theorem handles parsed.)

11.4 Complete Claim Coverage Matrix

Paper handle	Status	Lean support
cor:exact-identifiability	Derived	DQ.Sigma2PHardness.exactlyIdentifiesRelevant_iff_sufficient_and_subset_relevantFinset
cor:exact-no-competence-zero-certified	Full	DQ.IntegrityCompetence.certificationOverheadBits_of_no_evidence
cor:gap-externalization	Full	DQ.HardnessDistribution.simplicityTax_grows
cor:gap-minimization-hard	Derived	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
cor:generalized-eventual-dominance	Full	DQ.HardnessDistribution.generalized_right_eventually_dominates_wrong
cor:information-barrier-query	Full	DQ.ClaimClosure.exact_raw_eq_certified_iff_certainty_inflation_core, DQ.ClaimClosure.exact_raw_only_of_no_exact_admissible_core, DQ.ClaimClosure.explicit_assumptions_required_of_not_excused_core
cor:integrity-universal	Full	DQ.ClaimClosure.hard_family_all_coords_core
cor:linear-positive-no-saturation	Full	DQ.HardnessDistribution.no_positive_slope_linear_represents_saturating
cor:no-auto-minimize	Full	DQ.ClaimClosure.horizonTwoWitness_immediate_empty_sufficient
cor:no-uncertified-exact-claim	Full	DQ.ClaimClosure.information_barrier_value_entry_core

cor:outside-excuses-no-exact-report	Full	DQ.ClaimClosure.information_barrier_opt_oracle_core
cor:overmodel-diagnostic-implication	Derived	DQ.Sigma2PHardness.representationGap_eq_zero_iff
cor:physics-no-universal-exact-claim	Full	DQ.ClaimClosure.horizon_gt_one_bridge_can_fail_on_sufficiency
cor:practice-bounded	Full	agentBridgeClass
cor:practice-diagnostic	Derived	DQ.Sigma2PHardness.min_representationGap_zero_iff_relevant_card
cor:practice-structured	Full	agent_transfer_licensed_iff_snapshot
cor:practice-tree	Full	anchor_sigma2p_complete_conditional
cor:practice-unstructured	Full	ExactPhysicalClaimWellTyped
cor:query-obstruction-bool	Full	DQ.ClaimClosure.no_auto_minimize_of_p_neq_conp, OneStepSequentialObjective
cor:right-wrong-hardness	Full	DQ.HardnessDistribution.right_dominates_wrong
cor:type-system-threshold	Full	DQ.HardnessDistribution.native_dominates_manual
prop:abstention-frontier	Full	DQ.ClaimClosure.no_exact_claim_under_declared_assumptions_unless_excused_core
prop:adq-ordering	Full	DQ.ClaimClosure.adq_ordering
prop:attempted-competence-matrix	Full	DQ.IntegrityCompetence.EvidenceForReport, DQ.IntegrityCompetence.ExactCertaintyInflation, DQ.IntegrityCompetence.certaintyInflation_iff_not_admissible
prop:bridge-failure-horizon	Full	DQ.ClaimClosure.epsilon_admissible_iff_raw_lt_certified_total_core, DQ.ClaimClosure.exact_admissible_iff_raw_lt_certified_total_core
prop:bridge-failure-stochastic	Full	DQ.ClaimClosure.selectorSufficient_not_implies_setSufficient
prop:bridge-failure-transition	Derived	DQ.ClaimClosure.tractable_tree_core
prop:bridge-transfer-scope	Full	DQ.ClaimClosure.integrity_resource_bound_for_sufficiency
prop:budgeted-crossover	Full	DQ.ClaimClosure.meta_coordinate_not_relevant_on_declared_slice, DQ.PhysicalBudgetCrossover.explicit_infeasible_succinct_feasible_of_crossover
prop:certified-confidence-gate	Full	DQ.IntegrityCompetence.certificationOverheadBits, DQ.IntegrityCompetence.certificationOverheadBits_of_evidence
prop:crossover-above-cap	Full	DQ.ClaimClosure.meta_coordinate_irrelevant_of_invariance_on_declared_slice
prop:crossover-not-certification	Full	CC.lt
prop:crossover-policy	Full	DQ.ClaimClosure.minsuff_collapse_to_conp_conditional
prop:declared-contract-selection-validity	Full	DQ.ClaimClosure.declaredRegimeFamily_complete, DQ.ClaimClosure.information_barrier_opt_oracle_core
prop:dominance-modes	Full	DQ.HardnessDistribution.centralized_higher_leverage
prop:empty-sufficient-constant	Full	DQ.DecisionProblem.relevantSet_is_minimal
prop:evidence-admissibility-equivalence	Full	DQ.IntegrityCompetence.ReportSignal
prop:exact-requires-evidence	Full	DQ.IntegrityCompetence.ReportBitModel, DQ.IntegrityCompetence.SignalConsistent
prop:generalized-assumption-boundary	Full	DQ.HardnessDistribution.generalized_dominance_can_fail_without_right_boundedness, DQ.HardnessDistribution.generalized_dominance_can_fail_without_wrong_growth
prop:hardness-conservation	Full	DQ.HardnessDistribution.totalDOF_ge_intrinsic
prop:hardness-efficiency-interpretation	Full	DQ.HardnessDistribution.hardnessEfficiency_eq_central_share
prop:heuristic-reusability	Full	DQ.ClaimClosure.boundaryCharacterized_iff_exists_sufficient_subset, DQ.ClaimClosure.oracle_lattice_transfer_as_regime_simulation, DQ.ClaimClosure.physical_crossover_core
prop:identifiability-convergence	Full	DQ.ClaimClosure.exact_certainty_inflation_under_hardness_core
prop:insufficiency-counterexample	Full	DQ.ClaimClosure.DecisionProblem.sufficient_iff_zeroEpsilonSufficient, DQ.DecisionProblem.minimalSufficient_iff_relevant
prop:integrity-competence-separation	Full	DQ.IntegrityCompetence.Percent, DQ.IntegrityCompetence.admissible_matrix_counts

prop:integrity-resource-bound	Full	DQ.ClaimClosure.explicit_state_upper_core, DQ.IntegrityCompetence.abstain_signal_exists_with_guess_self, DQ.IntegrityCompetence.admissible_irrational_strictly_more_than_rational
prop:mdp-tractable	Full	DQ.ClaimClosure.query_obstruction_finite_state_core
prop:minimal-relevant-equiv	Full	DP.anchorSufficient, DQ.ClaimClosure.DecisionProblem.epsOpt_zero_eq_opt
prop:no-evidence-zero-certified	Full	DQ.IntegrityCompetence.certifiedTotalBits
prop:one-step-bridge	Full	DQ.ClaimClosure.integrity_resource_bound_for_sufficiency
prop:oracle-lattice-strict	Full	ExcusedBy, ExplicitHardnessAssumptions
prop:oracle-lattice-transfer	Full	DQ.ClaimClosure.integrity_universal_applicability_core, boundaryCharacterized
prop:outside-excuses-explicit-assumptions	Full	DQ.ClaimClosure.declaredRegimeFamily_complete
prop:physics-no-universal-exact	Full	DQ.ClaimClosure.cost_asymmetry_eth_conditional
prop:query-finite-state-generalization	Full	RegimeSimulation, adq
prop:query-randomized-robustness	Full	L.eval, TransitionCoupledObjective, TwoStepObjective
prop:query-randomized-weighted	Full	bounded_actions_reusable_heuristic, bridgeFailureWitness
prop:query-regime-obstruction	Full	DQ.ClaimClosure.no_exact_claim_admissible_under_hardness_core, RegimeSimulation, adq
prop:query-state-batch-lb	Full	StandardComplexityAssumptions, adq_ordering
prop:query-subproblem-transfer	Full	DQ.ClaimClosure.one_step_bridge, DQ.ClaimClosure.sufficiency_conp_complete_conditional, DQ.ClaimClosure.sufficiency_conp_reduction_core
prop:query-tightness-full-scan	Full	StructureDetectable
prop:query-value-entry-lb	Full	StochasticCriterionObjective, anchor_sigma2p_reduction_core
prop:query-weighted-transfer	Full	boundaryCharacterized_iff_exists_sufficient_subset, bounded_actions_detectable
prop:refinement-strengthens	Full	DQ.ClaimClosure.thermo_mandatory_cost_core
prop:retraction-evidence-integrity	Full	DQ.ClaimClosure.tractable_bounded_core
prop:retraction-no-evidence-violates	Full	DQ.ClaimClosure.tractable_separable_core
prop:selector-separation	Full	DQ.ClaimClosure.physical_crossover_above_cap_core
prop:sequential-bounded-horizon	Full	DQ.ClaimClosure.query_obstruction_boolean_corollary
prop:sequential-static-relation	Full	DQ.ClaimClosure.physical_crossover_hardness_core
prop:set-to-selector	Full	DP.constant_opt_all_sufficient
prop:snapshot-process-typing	Full	DQ.ClaimClosure.agent_transfer_licensed_iff_snapshot, DQ.ClaimClosure.minsuff_conp_complete_conditional, DQ.ClaimClosure.regime_core_claim_proved
prop:static-stochastic-strict	Unmapped	*(no derived Lean handle found)*
prop:static-stochastic-transfer	Full	DQ.ClaimClosure.regime_simulation_transfers_hardness
prop:stochastic-bounded-support	Full	DQ.ClaimClosure.subproblem_hardness_lifts_to_full
prop:stochastic-product-tractable	Full	DQ.ClaimClosure.subproblem_transfer_as_regime_simulation
prop:stochastic-sequential-bridge-fail	Full	DQ.ClaimClosure.snapshot_vs_process_typed_boundary
prop:stochastic-sequential-strict	Unmapped	*(no derived Lean handle found)*
prop:sufficiency-char	Full	DQ.ClaimClosure.thermo_energy_carbon_lift_core, DQ.ClaimClosure.thermo_eventual_lift_core
prop:thermo-conservation-additive	Full	DQ.ClaimClosure.tractable_subcases_conditional
prop:thermo-hardness-bundle	Full	DQ.ClaimClosure.tree_structure_detectable

prop:thermo-lift	Full	DQ.ClaimClosure.tractable_tree_core, DQ.ClaimClosure.transition_coupled_bridge_can_fail_on_sufficiency
prop:thermo-mandatory-cost	Full	DQ.ClaimClosure.typed_claim_admissibility_core
prop: typed-claim-admissibility	Derived	DQ.ClaimClosure.tree_structure_detectable
prop: universal-solver-framing	Derived	DQ.ClaimClosure.typed_static_class_completeness
prop: zero-epsilon-competence	Full	DQ.IntegrityCompetence.RLFFWeights, DQ.IntegrityCompetence.certifiedTotalBits_ge_raw
prop: zero-epsilon-reduction	Full	DP.classMonotoneOn, DQ.DecisionProblem.sufficient_implies_selectorSufficient
thm:amortization	Full	DQ.HardnessDistribution.complete_model_dominates_after_threshold
thm:bridge-boundary-represented	Full	DQ.ClaimClosure.bounded_actions_detectable, DQ.ClaimClosure.bridge_boundary_represented_family, DQ.ClaimClosure.bridge_failure_witness_non_one_step
thm: centralization-dominance	Full	DQ.HardnessDistribution.centralization_dominance_bundle, DQ.HardnessDistribution.centralization_step_saves_n_minus_one
thm:claim-integrity-meta	Full	DQ.ClaimClosure.declaredRegimeFamily_complete
thm:config-reduction	Full	DQ.ConfigReduction.config_sufficiency_iff_behavior_preserving
thm:cost-asymmetry-eth	Full	DQ.ClaimClosure.bridge_transfer_iff_one_step_class, DQ.HardnessDistribution.linear_lt_exponential_plus_constant_eventually
thm:dichotomy	Full	DQ.ClaimClosure.declaredBudgetSlice, DQ.ClaimClosure.declared_physics_no_universal_exact_certifier_core, DQ.ClaimClosure.dichotomy_conditional
thm:generalized-dominance	Full	DQ.HardnessDistribution.generalized_right_dominates_wrong_of_bounded_vs_identity_lower
thm:generalized-saturation-possible	Full	DQ.HardnessDistribution.generalizedTotal_with_saturation_eventually_constant, DQ.HardnessDistribution.saturatingSiteCost_eventually_constant
thm: linear-saturation-iff-zero	Full	DQ.HardnessDistribution.totalDOF_eventually_constant_iff_zero_distributed
thm:overmodel-diagnostic	Full	DQ.ClaimClosure.anchor_sigma2p_reduction_core, DQ.ClaimClosure.information_barrier_state_batch_core
thm:regime-coverage	Full	DQ.ClaimClosure.certified_total_bits_split_core, DQ.ClaimClosure.no_exact_identifier_implies_not_boundary_characterized
thm:tax-conservation	Full	DQ.HardnessDistribution.gap_conservation_card
thm:tax-grows	Derived	DQ.HardnessDistribution.totalDOF_ge_intrinsic
thm:tractable	Full	DQ.ClaimClosure.universal_solver_framing_core
thm: typed-completeness-static	Derived	DQ.ClaimClosure.typed_claim_admissibility_core

Notes: (1) Full rows come from theorem-local inline anchors in this paper. (2) Derived rows are filled by dependency/scaffold claim-handle derivation (same paper-handle label across proof dependencies). (3) Unmapped means no local anchor and no derivable dependency support were found.
Auto summary: mapped 100/102 (full=91, derived=9, unmapped=2).

11.5 Claims Not Fully Mechanized

Entries marked **Full (conditional)** are explicitly mechanized transfer theorems that depend on standard external complexity facts (e.g., MAJSAT PP-completeness, TQBF PSPACE-completeness), with those dependencies represented as hypotheses in Lean.

11.6 Verification

The proofs compile with Lean 4. Run `lake build` in the proof directory to verify.

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